Set-Asides Reshape Bids, Not Bidders: Evidence from Brazilian Medical Procurement*

Darcio Genicolo-Martins†

April 2026

Abstract. Does a small-firm set-aside raise prices by excluding competitors, or by changing how the remaining firms bid? For Brazil's March 2018 extension of the SME-only rule to medical supplies, the Campo et al. (2003) estimator on 33,706 auctions decomposes the 10–11% price effect: **66% is within-auction bidding response, 34% is cost-primitive entry.** A testable primitive-invariance condition for non-SMEs holds tightly (KS $D = 0.03$). Participation shifts are large ($-62\%/+52\%$), but the price effect lives inside the auction, not at the door. Deadweight loss is 11.8% of procurement value; a top-quartile exemption recovers 95% of the fiscal cost. **Set-asides reshape bids, not bidders.**

JEL codes: H32, H57, L26, L53, D44, C14.
Keywords: public procurement; small-firm set-asides; structural auction estimation; asymmetric GPV; welfare counterfactuals.

**This version: April 2026 (v2-structural).* The structural complement to the reduced-form evidence in Genicolo-Martins (2026a); the two papers should be read in sequence. Replication materials: https://github.com/darciogm/paper2-me-epp/v2-structural. All errors are my own.

†INSPER, São Paulo, Brazil. Email: darciogm1@insper.edu.br.

1 Introduction

When a government restricts public procurement to small firms, does the price premium come from the firms it excludes, or from the firms it keeps? The question is old—set-aside rules have been on the books in developed and developing economies since at least the 1950s—and the answer determines what reform looks like. If the premium runs through the extensive margin (fewer firms bid), the policy fix is to expand the small-firm roster: subsidies, registration relief, capacity-building. If instead it runs through the intensive margin (the same firms bid less aggressively because they no longer face large competitors), no roster expansion will close the gap—the only lever is to shrink the *scope* of the rule itself. Policy prescriptions that differ by an order of magnitude turn on a quantity the literature has mostly been unable to measure directly.

This paper measures it, using a quasi-experiment in Brazilian medical procurement. Between August 2014 and February 2018, medical supplies (Group 65 in the BEC product catalog) were the *only* major product group in São Paulo state that remained open to general competition; 76 other product groups had been under the SME-only tendering rule since the federal 2014 mandate took effect. The exception was the outcome of a joint agreement between the state government and its audit court (TCE-SP) that medical supplies required open competition on public-health grounds. That exception ended in March 2018, when a legal opinion from a separate state agency (the *Procuradoria-Geral do Estado*, PGE-SP) held the exemption unconstitutional on isonomy grounds: medical supplies could not be procedurally distinguished from other groups. The SME-only rule extended to Group 65 overnight. The reversal was triggered by a bureaucratic legal argument, not by any procurement-market outcome; it is plausibly exogenous to anything happening in Group-65 bidding.

The approach. Section 3 establishes via difference-in-differences that the policy raises Group-65 procurement prices by 10–11% over the 18-month window, a figure robust to the Callaway and Sant'Anna (2021), Sun and Abraham (2021), and Goodman-Bacon (2021)

estimators. Reduced-form mediation, however, cannot separate the two channels of interest. A Gelbach (2016) decomposition locates roughly 95% of the effect outside the observable entry and winner-composition channels. That residual is what the present paper explains: using bid distributions recovered from the BEC electronic platform, I estimate an asymmetric first-price sealed-bid (FPSB) model on the *Convite* modality of Group-65 procurement (33,706 auctions with at least two bidders; 24,346 bidder-auctions entering the CPV inversion) via the Campo et al. (2003) extension of Guerre et al. (2000). The estimation delivers type-specific cost distributions $F_c^{\text{SME}}(\cdot)$ and $F_c^{\text{NonSME}}(\cdot)$. Because the regime switch affects *which* firms bid but not their primitive costs, the non-SME cost distribution should be invariant across the break. That invariance, when it holds, both validates the model and pins down the channel structure.

Three findings. *First*, the identifying restriction holds where the model predicts it should. In Convite, where non-SMEs continue bidding in many auctions both before and after the cutoff, their recovered cost distribution is statistically invariant across the March-2018 break: mean shift of 0.001 of reference price, KS $D = 0.03$. Under the Pregão Haile-Tamer upper bound, the pattern flips—it is the SME bound that is nearly invariant ($D = 0.03$), while the non-SME bound drifts upward by 0.04 of reference. The drift reflects selection among the shrinking pool of non-SMEs who keep bidding despite near-zero post-policy win probability, not a movement in primitive costs. In each modality, the primitive that survives the test is the one whose bidder pool was not re-selected by the policy.

Second, the SME cost distribution does shift upward—by 7.7% relative to its Pre-period mean, equivalent to 4 percentage points in ref-normalized units (KS $D = 0.08$, $p < 10^{-10}$). This is the compositional footprint of the policy: when Group 65 becomes SME-only, marginal higher-cost small firms—firms that previously specialized in other product groups or new entrants—enter the Group-65 candidate pool. The SME primitive is no longer the same pool before and after. This "entry margin" is, to my knowledge, the first structural measurement

of pool re-composition under an SME set-aside.

Third, the structural price effect decomposes as **∼66% intensive margin and ∼34% entry margin**. The intensive share is 74% at N=2, 47% at N=3, 70% at N=4, and 72% at $N \geq 5$—a rough two-thirds across auction sizes. This is the central structural finding. It does the work the Gelbach residual could not: the 95% unexplained by observable entry and winner composition is, at its core, *within-auction bidding behavior*—the same SMEs bidding differently once their larger competitors are gone—not another observable margin waiting to be added. Participation responds too, and substantially: the reduced-form entry regression shows the non-SME entry rate falls 62% under the restriction ($t = 43$) while the SME entry rate rises 52% ($t = 38$). These shifts amplify the total welfare cost the policy generates, but they are not where the price effect primarily lives. The roughly R$30–33 (US$9–10) type-symmetric entry cost implied by the free-entry zero-profit condition means that a rigorous Athey et al. (2011) nested entry-bidding model— future work—should refine magnitudes without overturning the intensive-margin primacy of the fixed-pool result.

Welfare counterfactual. Using the recovered cost distributions, I simulate equilibrium procurement prices under a *value-threshold exemption*— applying the SME-only rule only below a reference-price cutoff. The structural Pareto frontier is strikingly concave: exempting the top quartile of items by reference price preserves the SME preference on 75% of items *by count* while recovering **95%** of the fiscal cost of the uniform rule. This reform is not just arithmetic—the structural model shows that the top-quartile items are where the intensive margin bites hardest, so reopening them is also where cost recovery is largest.

Contributions. The findings above distill into three methodological contributions to the literature:

(i) First structural decomposition of an SME set-aside into intensive and entry channels. The public-procurement literature has either measured SME effects in reduced form (Marion, 2007; Denes, 1997; Ferraz et al., 2015) or modeled bid preferences structurally

without the clean *restriction*-versus-*preference* contrast (Krasnokutskaya and Seim, 2011a; Athey et al., 2013). I combine the two: reduced-form identification via a discrete legal cutoff, followed by structural estimation of type-specific cost primitives. The headline finding—that within-auction bidding carries the majority of the policy's price effect—is directly relevant to the design of SME preference regimes in Brazil, the EU, and the 130 other jurisdictions documented in OECD (2018); SIGMA (2016); Hoekman and Tas (2020).

(ii) A selection-conditional consistency check on primitives. In the Convite CPV estimator, the non-SME cost distribution is tightly identified across the regime break: it enters a (same) set of auctions with (essentially) the same bidder-pool structure before and after the policy. The resulting KS $D = 0.03$ (mean shift 0.001) is a consistency check on the model under the identifying assumption that the policy did not re-select which non-SMEs bid in the Convite Pre vs Post samples. This condition plausibly holds in Convite because non-SMEs remain eligible to bid in Convite auctions not subject to the SME-only restriction, and the regime change affects a subset of Group-65 auctions rather than the entire non-SME choice set. Under Pregão, where the Haile-Tamer upper bound is applied to final bids and non-SMEs continue to bid despite near-zero post-policy win probability, the upper bound on the non-SME cost distribution does shift ($D = 0.06$), consistent with selection of the remaining non-SME bidders toward higher-cost types. The paper's identification relies on the Convite cross-regime check, with the Pregão result as a directional robustness.

(iii) Welfare-based structural evaluation of a value-threshold reform. I simulate a concrete policy alternative —exempting high-reference-price items from the SME-only rule while keeping the preference on the rest—and show that the structural primitives deliver a Pareto frontier considerably more concave than an accounting-identity calculation would suggest. The top-quartile exemption recovers 95% of the fiscal cost on 25% of items, grounded in asymmetric-bidder welfare accounting rather than mechanical extrapolation.

Relation to the literature. The paper connects two strands. The empirical SME procurement literature—Marion (2007); Denes (1997); Ferraz et al. (2015); Szucs (2024)—documents price and efficiency effects of set-aside and preference rules in reduced form. The structural auction literature—Guerre et al. (2000); Campo et al. (2003); Krasnokutskaya and Seim (2011b); Athey et al. (2013); Bajari et al. (2014); Bajari and Hortaçsu (2005); Hendricks et al. (2003)—recovers cost primitives from bid data and simulates counterfactuals, typically in settings without a clean regime break. This paper bridges the two: the March 2018 break provides the causal cut; CPV asymmetric estimation recovers the primitives; and the structural counterfactuals quantify welfare. The approach is closest in spirit to Krasnokutskaya and Seim (2011a) on California highway procurement with bid preferences, but with three distinctions: (i) the regime contrast is *restriction* vs *open*, not a 5% preference; (ii) the asymmetric bidder-type framework is Campo et al. (2003) rather than symmetric GPV with unobserved heterogeneity as in Krasnokutskaya and Seim (2011b); and (iii) the March 2018 regime break provides a *testable* primitive-invariance condition on the non-SME cost distribution, which Krasnokutskaya and Seim (2011a) assume rather than test.

For comparison of magnitudes: Krasnokutskaya and Seim (2011a) analyze 697 Caltrans projects with 3,034 bids, an average of 4.4 bidders per project (1.75 small + 2.62 large). Their counterfactual estimates the California 5% Small Business Preference raises procurement cost by 0.5% and shifts small firms' share of work from 12.5% to 15.6%; a 25% bid discount would raise costs by 4.1% (their Table 11). The present paper's set-aside regime (complete exclusion of non-SMEs) is an order-of-magnitude stronger policy and generates a proportionally larger effect: a 10.4% fiscal cost, 11.8% DWL of procurement value, roughly 20× the Krasnokutskaya and Seim (2011a) 5% baseline. That ratio is broadly consistent with moving from a mild preference to a full restriction. Type-specific cost asymmetries are broadly consistent: Krasnokutskaya and Seim (2011a) estimate small firms bid 5.4% (parametric) to 8.1% (OLS) higher than large firms; we estimate small firms bid about 17% higher under the Pregão Haile-Tamer upper bound (H1-only; one-sided bound), about 4.7% higher under the canonical

Convite CPV (Appendix C), and essentially zero under the proxy-classification Convite CPV (where the proxy pools firms by registry class and the cross-type distinction is noisier at the margin).

Roadmap. Section 2 describes the institutional background—the Brazilian procurement regime, the BEC platform, and the 2014–2018 regulatory history of Group 65 that motivates the identifying cut. Section 3 summarizes the reduced-form baseline establishing the 10–11% headline price effect and the 95% Gelbach residual that the structural exercise will decompose. Section 4 presents the asymmetric FPSB procurement model. Section 5 states the identification strategy and the Campo-Perrigne-Vuong inversion. Section 6 describes the bid-level data and the Convite modality. Section 7 estimates the model and reports the primitive-invariance tests. Section 8 delivers the headline 66/34 decomposition of the structural price effect into intensive and entry channels. Section 9 documents the participation response (62% non-SME exit, 52% SME entry) and shows why it does not overturn the within-auction primacy of the intensive margin. Section 10 presents the welfare analysis—accounting identity, DWL magnitudes, annualization, and the value-threshold counterfactual. Section 11 discusses robustness (Pregão type-split Haile-Tamer bounds, canonical SME, IPV-vs-CV diagnostic). Section 12 concludes.

2 Institutional Background

The identifying variation in this paper starts with a single bureaucratic document. In early 2018, São Paulo's state attorney's office (PGE-SP) issued an internal legal opinion that, on grounds entirely unrelated to procurement outcomes, extended Brazil's SME-only default to its last open product category: medical and hospital supplies. A parecer on equal treatment took effect the following month, and the auctions in this paper look different on either side of that line. To see why the line is clean, it helps to understand how the surrounding system works—the statutory layer-cake of Brazilian procurement, the BEC platform that

operationalizes it in São Paulo, the 2014 federal mandate that made SME-only tendering binding, the peculiar 2014–2018 exception that kept Group 65 outside the rule, and the 2018 reinterpretation that swept it in.

Brazilian public procurement: statutory framework. Brazilian procurement is a layered structure, and each layer leaves a trace in the data. The foundational statute is Lei 8.666/1993, which at the time of this study defined five classical modalities sorted by value and formality: *Concorrência* for the largest projects, *Tomada de Preços* for pre-registered suppliers, *Convite* for invited sealed-bid auctions, *Concurso* for creative work, and *Leilão* for the sale of public assets. Lei 10.520/2002 added a sixth modality, the *Pregão*—a reverse auction that opens with sealed proposals and closes with an iterative open-outcry phase—and over time made it the workhorse of standardized procurement. Convite and Pregão are the two modalities this paper uses: Convite because it is a first-price sealed-bid auction, exactly the object the CPV estimator was designed for, and Pregão because its iterative structure puts it in the Haile-Tamer world of ascending-bid bounds. Two additional pieces of legislation matter for timing: Decreto 9.412/2018 updated the modality ceilings mid-sample (raising the Convite non-engineering item limit from R$80,000 to R$176,000), and Lei 14.133/2021 is currently phasing in a unified replacement for 8.666 and 10.520. Neither change affects the identifying window in this paper, but both matter for external validity.

Layered on top of these modalities is a separate preferences regime for small firms, introduced by Lei Complementar 123/2006—the Microenterprise Statute, known in practice as LC 123. LC 123 defines two eligible firm sizes—*Micro-empresa* (ME: annual revenue $\leq$ R$360,000) and *Empresa de Pequeno Porte* (EPP: $\leq$ R$4.8M)—and grants them three cumulative preferences: exclusive eligibility for items below R$80,000 (Art. 48 I), subcontracting set-asides (Art. 48 II), and a 10% tie-breaking margin, reduced to 5% in Pregão (Art. 44 §2). For the first eight years of the statute these preferences were optional tools that purchasing units could invoke or ignore. That changed in 2014: LC 147/2014 converted

them into binding defaults, requiring any buyer who ran a general (non-restricted) tender for a sub-R$80,000 item to file a written justification with the state audit court. The 2014 amendment is the proximate source of the SME-only regime this paper studies—and the reason the Group-65 exception that followed was such an anomaly.

BEC: the São Paulo procurement platform. The statutes describe rules; BEC enforces them. Since Decreto estadual 47.297/2002, every São Paulo state purchase passes through the *Bolsa Eletrônica de Compras*, an electronic procurement platform administered by the state treasury secretariat (SEFAZ/SP). BEC is the single point of entry for 1,344 purchasing units—state bureaus, participating municipalities, and accredited private entities subject to state rules—and in 2019 it moved roughly R$13 billion (US$3.7 billion) through its order book. Cumulatively since 2005, BEC has handled more than R$105 billion (US$30 billion), spread across 860,000 purchase offers and 4.8 million item-level transactions. That volume alone makes the platform an unusually dense laboratory for procurement research.

Three nested registries structure the platform. Suppliers enroll in the Cadastro Unificado de Fornecedores (CAUFESP), which certifies their tax regularity, firm size, and procurement eligibility; goods and services are cataloged in a three-tier CADMAT taxonomy (*group* $\rightarrow$ *class* $\rightarrow$ *item*); and each auction runs under one of the statutory modalities, with Pregão absorbing roughly 70% of value, Convite 15%, and Dispensa eletrônica (below-R$17,600 direct-purchase auctions) 15%. In scope BEC parallels the federal ComprasNet, but it has its own governance, its own supplier certification, and a different closing protocol—BEC's Pregão uses a soft-close rule rather than the "tempo aleatório" randomization that ComprasNet adopted. For this paper the relevant CADMAT address is Group 65 (medical and hospital supplies), and within it class 6531, which covers pharmaceutical medications subject to additional ceiling regulation by the Câmara de Regulação do Mercado de Medicamentos (CMED). The three-letter acronyms matter: CAUFESP defines who can bid, CADMAT defines what is being bid on, and CMED caps what some of it can cost.

Enforcement of the SME-only rule. What LC 147 converted from an option into a default, TCE-SP is what turns the default into behavior. Enforcement in BEC operates through *opt-out costs*: a purchasing unit that wants to run a general (non-restricted) tender for an eligible item must file a formal justification with the Tribunal de Contas do Estado—the state audit court, and a body with real teeth. The filing is reviewed, and a unit found to have circumvented the SME-only rule without adequate basis can face administrative sanctions, reporting penalties, and loss of procurement autonomy. Buyers notice. State-level adherence to SME-only tendering for eligible items rose from roughly 13% before LC 147 to close to 70% within three years, and by 2017 an un-justified general tender on a sub-threshold item had become unusual enough to draw scrutiny on its own. Buyers keep their files clean by using the rule, not by arguing with it.

The R$80,000 value ceiling is only half the trigger. Art. 49 of LC 123 adds a second condition: at least three eligible SMEs must be "potentially available" for the item. Fewer than three, and the auction falls back to the general rule. This *three-SMEs precondition* creates a natural item-level fault line that is partly endogenous to the item's market. It also becomes the lever of the 2014–2018 exception described next.

The Group-65 exception (2014–2018). One product group did not look like the others. In the months after LC 147 took effect in 2014, São Paulo's state government and TCE-SP together argued that medical and hospital supplies belonged in a different regulatory category—"strategic goods" that warranted open competition on public-health grounds—and the audit court issued a resolution carving Group 65 out of the SME-only default. The argument rested on three supporting points. First, medical procurement has unusual downstream stakes: the difference between an adequately stocked hospital and a stock-out is not the same as the difference between two office supplies, and rigorous price discovery mattered. Second, ANVISA's pharmaceutical registration system narrows the qualified-supplier pool for many medical items below the three-SMEs threshold of LC 123, which on this reading would

have made the restriction inapplicable to large shares of the group anyway. Third, CMED's regulated list prices for class-6531 medications already impose a price ceiling that constrains buyer welfare, making the additional SME restriction redundant for the affected items. The three arguments together left Group 65 uniquely outside the SME-only regime from August 2014 through February 2018—the sole major product group so exempted during the period, and, on the data, a regime-free control that the rest of the catalogue did not have.

The March 2018 reversal. The 2014 carve-out was a compromise between the buying side (SEFAZ/SP) and the audit side (TCE-SP). The 2018 reversal came from a third actor—the *Procuradoria-Geral do Estado* (PGE-SP), the state's constitutional and administrative legal counsel—which had not been at the original table. In November 2017, in the course of a routine internal legal review, PGE-SP issued a formal opinion (parecer) concluding that the Group-65 exception violated *isonomia*, the equal-treatment principle codified in Art. 5 of the Federal Constitution. The opinion did not engage with procurement outcomes, price trajectories, or supplier complaints; it argued from first principles that procedural symmetry across product groups required a substantive distinguishing feature, and found that the 2014 rationale did not clear that bar. ANVISA's registration system, the opinion held, operates at the item level and is already addressed by the three-SMEs precondition built into LC 123; CMED's price controls operate as a ceiling rather than a competitive floor, leaving the question of allocative efficiency untouched. The remedy, PGE-SP directed, was to extend the SME-only default to Group 65 on the same footing as every other group. The parecer was transmitted to BEC in early 2018 and took effect in March of that year. From the bidders' perspective, a category that had been open for more than three years became, overnight, another line in the SME-only catalogue.

Why this is a clean natural experiment. Three features of the March 2018 reversal support the identifying assumption that the cut is exogenous to the procurement process itself. First, the *trigger* was a legal reinterpretation of a constitutional principle, not a response to

market outcomes: no procurement-data audit, political event, or supplier complaint preceded the opinion, and the parecer cites none. Second, the *timing* was bureaucratic. PGE-SP's opinion was issued on its own internal schedule, transmitted to BEC in early 2018, and took effect in March 2018 without the sort of public consultation or draft-rule window that would have let bidders re-price in advance. Third, the *boundaries* of the experiment are legally clean: the affected set is a single well-defined product group (Group 65, meeting the three-SMEs precondition), and the unaffected controls are the other 76 product groups that had been operating under the SME-only default since 2014. An 18-month window straddling the break yields a crisp treated-vs-never-treated comparison.

These features motivate the identifying strategy that follows in Section 5. The reversal should leave the non-SME cost primitive alone; only the set of bidders that participate changes, not who any given firm is. That invariance is the core identifying assumption of this paper, and—crucially for what follows—it is testable directly against the data.

3 Reduced-Form Context

Before turning to primitives, two reduced-form facts motivate the structural exercise: the March 2018 regime switch causes a 10–11% price increase, and a Gelbach mediation decomposition locates *95%* of it outside any observable channel. This section establishes both.

Design and estimate. March 2018 places Group 65 (the treated cohort) under the SME-only default that had already applied to 76 other product groups since 2014. Identification proceeds via a staggered-DiD reversal: Group 65 switches from "open" into "SME-only" while the 76 comparison groups remain SME-only throughout an 18-month window. The Goodman-Bacon (2021) decomposition places 100% of the two-way-fixed-effects weight on the clean treated-vs-never-treated 2×2 comparison—there is nothing to re-aggregate. The

main regression is

$$y_{ipt} \;=\; \beta\,(g65_i \times Pre_t) \;+\; \zeta_i \;+\; \mathbf{X}'_{ipt}\gamma \;+\; \varepsilon_{ipt}, \tag{1}$$

with item fixed effects ζ_i, controls $\mathbf{X}$ including a sealed-bid indicator and log quantity, and standard errors clustered at the item level. Table 1 reports $\hat{\beta}$ across four outcomes and three windows.

Table 1. Reduced-form headline DiD estimates (Group 65 vs 76 controls)

	6-month		12-month		18-month	
	Baseline	+ PBU FE	Baseline	+ PBU FE	Baseline	+ PBU FE
Log winning price	−0.142	−0.148	−0.108	−0.108	−0.109	−0.113
	(0.015)	(0.014)	(0.013)	(0.013)	(0.012)	(0.012)
Log firm count	+0.211	+0.220	+0.128	+0.138	+0.098	+0.107
	(0.009)	(0.009)	(0.007)	(0.008)	(0.007)	(0.007)
Log valid bids	+0.203	+0.215	+0.103	+0.114	+0.069	+0.080
	(0.013)	(0.014)	(0.010)	(0.011)	(0.010)	(0.011)
Winner-PBU dist. (km)	+6.64	+1.94	+13.78	+5.70	+14.25	+6.15
	(3.32)	(1.98)	(3.02)	(2.38)	(3.08)	(2.22)

Notes: Each cell is $\hat{\beta}$ (SE) from Equation (1) for the outcome-window-specification combination. Item fixed effects included throughout. "Baseline" columns include sealed-bid and log-quantity controls; "+ PBU FE" columns add buyer-unit fixed effects. SE clustered at the item level. Sample sizes vary by outcome and window (∼250k–700k item-level observations). All price and firm-count coefficients are significant at $p < 0.01$. Own calculation on BEC bid-level data; replication scripts in the supplementary materials.

Four patterns emerge from the 18-month baseline. **Winning prices rise 10–11%** (using $\exp(\hat{\beta}) - 1$; sharper in shorter windows). **Firm count falls 10–11%** despite a policy mandating SME participation—non-SMEs exit faster than SMEs enter. **Valid bids fall only 6–8%**, consistent with each remaining SME submitting more bids per auction than the non-SMEs they replaced. **Winner-to-PBU distance falls 5–14 km**, consistent with SMEs being more geographically local. The price coefficient is stable across the Callaway-Sant'Anna, Sun-Abraham, and Goodman-Bacon heterogeneity-robust estimators (−0.108 to −0.114); cluster-robust to item-group, PBU, and two-way clustering; wild cluster bootstrap delivers $p < 0.001$; a synthetic-control comparison confirms the post-treatment gap; a formal spillover test bounds implied bias at +0.0005 log-points; a Rambachan and Roth (2023) HonestDiD diagnostic rules out plausible parallel-trends violations.

Gelbach (2016) decomposition: the puzzle. The natural first question is whether the price jump is compositional—fewer bidders, different types. A Gelbach (2016) decomposition of the price regression adds log firm count and a winner-SME indicator as mediators, asking how much of $\hat{\beta}$ they jointly explain. The answer is striking: *almost none of it.* The two mediators move in opposite directions—log firm count pulls prices down by +181% of the headline, winner-SME composition pulls them up by −81%—and what looks like a large decomposition is a near cancellation. The net mediated share is about 5%; the residual—the share of the price effect that observable entry and observable composition cannot explain—is about **95%**. That residual is the puzzle the rest of the paper addresses.

From reduced form to structural. The Gelbach cancellation rules out easy compositional stories. What remains must be one (or more) of: (i) within-auction bidding behavior that shifts with the candidate pool, (ii) unobserved compositional shifts, or (iii) selection of firms into bidding with different cost primitives. The reduced form cannot separate these; the CPV inversion can. The rest of the paper pulls the 95% residual apart into a within-auction intensive margin (∼66%), a cost-primitive entry margin (∼34%), a documented participation response that amplifies the policy's reach without driving the price effect, and a welfare calculation in both R$ and US$. Extended reduced-form analysis—placebos, spillover bounds, policy-design variants— is developed in a parallel paper (Genicolo-Martins, 2026); the present paper is self-contained on all quantities it uses. The reduced form tells us where to look; the structural model tells us what we are looking at.

4 A Model of Asymmetric FPSB Procurement with Type-Specific Costs

Primitives. The environment is a first-price sealed-bid reverse (procurement) auction for a single indivisible good, with two bidder types that match the institutional distinction the

policy draws: type-A (*SME*) and type-B (*non-SME*). In auction t, there are $n_A^t \in \{0, 1, 2, \ldots\}$ SMEs and $n_B^t \in \{0, 1, 2, \ldots\}$ non-SMEs. Type-k bidders draw private costs $c \sim F_k$ i.i.d. of each other and of auction characteristics. The distributions F_A and F_B are continuous with densities f_A, f_B on $[\underline{c}, \overline{c}]$. The two type-specific distributions are the objects the paper is ultimately trying to recover from the data, and the separation between them is what allows the SME-only policy to be a policy rather than a relabeling.

Each bidder observes only her own cost and the auction's type composition (n_A^t, n_B^t). Each submits a sealed bid $b \in [0, \overline{c}]$; the lowest bid wins; ties are broken by a fair lottery. The payoff to a type-k bidder with cost c who bids b is $(b - c) \cdot \mathbf{1}\{b \text{ wins}\}$.

Equilibrium. In a symmetric-within-type Bayes-Nash equilibrium, each type-k bidder uses the same bidding function $\beta_k : [\underline{c}, \overline{c}] \to \mathbb{R}_+$, which is strictly increasing and continuously differentiable on the interior of its support. Let $G_k(b) = \Pr(b_k \leq b)$ denote the induced bid distribution and $g_k(b)$ its density. These are observables from auction data.

A type-k bidder with cost c maximizes

$$\pi_k(b, c; n_A^{-k}, n_B^{-k}) = (b - c) \cdot \Big[1 - G_A(b)\Big]^{n_A^*} \Big[1 - G_B(b)\Big]^{n_B^*},$$

where $n_A^* = n_A - \mathbf{1}\{k = A\}$, $n_B^* = n_B - \mathbf{1}\{k = B\}$ denote the *opponent* counts of each type.

The first-order condition is

$$b - c \;=\; \frac{1}{n_A^* \dfrac{g_A(b)}{1 - G_A(b)} + n_B^* \dfrac{g_B(b)}{1 - G_B(b)}}, \tag{2}$$

and the inverse equilibrium bid function, central to identification, is

$$c \;=\; b \;-\; \frac{1}{n_A^* \dfrac{g_A(b)}{1 - G_A(b)} + n_B^* \dfrac{g_B(b)}{1 - G_B(b)}}. \tag{3}$$

Equation (3) is the asymmetric extension of Guerre et al. (2000) developed by Campo et al. (2003), and it is the single piece of the model that does the estimation work. The symmetric-within-type special case ($G_A = G_B = G$, $g_A = g_B = g$) collapses to the familiar $c = b - (1 - G(b))/((N-1)g(b))$, the standard procurement GPV. Crucially, once type-specific bid distributions are estimated from the data, the inverse map is deterministic: there are no parameters to search over, no auxiliary distributions to assume. Whatever asymmetry exists between SME and non-SME costs shows up directly in the pseudo-costs.

Ref-price normalization. A practical obstacle is that reference prices P^t_{ref} span two orders of magnitude across Group-65 auctions, from small consumables to large-volume medication purchases. Running CPV on raw bids would mix scales that are not, in any economic sense, comparable. Following the bid-preference literature (e.g., Athey et al., 2013), I normalize each bid by its auction-specific reference price, $\tilde{b}^t = b^t/P^t_{\text{ref}}$, placing all normalized bids on a common $[0, 2]$ scale. Equation (3) then applies to $\tilde{b}$, with $\tilde{c} = c/P_{\text{ref}}$ on the same scale. The normalization amounts to an assumption that the cost distribution scales linearly with the buyer's reference price—a strong but standard assumption, and one that the invariance test in Section 7 provides an indirect check on. Level results without normalization are reported in the appendix.

5 Identification

Point identification from bid data. The CPV inversion is, at root, a one-line recipe. The model implies that G_A, G_B, g_A, g_B are directly observable from type-separated bid data, and that is what the bid-level records on BEC provide. Given these four objects and an auction's type composition (n^t_A, n^t_B), Equation (3) returns a pseudo-cost $\hat{c}_{i,t}$ for every bidder i in every auction t. Repeating the operation across all bidder-auctions delivers type-specific empirical cost distributions $\hat{F}_A$ and $\hat{F}_B$. Nothing more is needed: there is no nonlinear search, no auxiliary distributional assumption, no second estimation stage.

Two conditions have to hold for the recipe to be meaningful. First, asymmetric bidders must actually appear in the same auctions, so that both G_k are estimable on a non-degenerate sample. Convite Group 65 satisfies this condition comfortably in the Pre period—40% of auctions mix SMEs and non-SMEs. Second, bid functions must be monotonic in cost within type, which the model guarantees and which the data confirm (the pseudo-costs satisfy $\hat{c} \leq b$ on every observation in the estimation sample).

Identification check: primitive invariance. The policy gives us a second handle that most structural auction papers do not have: a testable over-identifying restriction. In the model, F_A and F_B are *primitives*—they describe the cost process before any equilibrium considerations enter. The March 2018 reversal changes *which* non-SMEs are eligible to bid in Group 65 (none, post-policy), but it cannot reach inside any given firm and change its cost draw. Therefore,

$$F_B^{\text{Pre}}(c) \;=\; F_B^{\text{Post}}(c) \quad \text{for all } c,$$

and a Kolmogorov-Smirnov test of exactly this equality is the paper's primary identification check. No analogous condition holds for the SME distribution: the policy changes the return to entering Group 65 for small firms, and we expect F_A^{Post} to shift relative to F_A^{Pre} precisely because marginal SMEs enter (or exit) under the restriction. Selection into the SME pool is part of the entry channel we want to measure. The over-identifying restriction is therefore one-sided by design: F_B invariant (testable), F_A allowed to move.

Robustness to mis-specified equilibrium. The price of using an equilibrium condition to recover primitives is that the recovery is only as good as the equilibrium. If bidders are risk-averse rather than risk-neutral, or if the auction has a common-value component, the pseudo-costs returned by Equation (3) are not consistent estimators of the underlying cost distribution. The invariance test on F_B is therefore a joint test of the model *and* of primitive invariance: passing it is evidence for both. Two useful features soften this concern in practice.

First, if the mis-specification is common to both bidder types, the invariance condition still identifies a weaker but coherent hypothesis about the unchanged non-SME pool. Second, the risk-aversion and common-value concerns are addressed directly in Section 11 using a CARA CGPV calibration and the Hendricks et al. (2003) diagnostic, respectively. Results in Section 7 suggest the model is reasonably specified: invariance holds tightly for non-SMEs (the type whose pool is supposed to be unchanged), and fails in exactly the direction the entry channel predicts for SMEs.

6 Data

BEC bid-level records. Because every São Paulo state purchase passes through a single electronic platform, the data requirements of a structural auction paper sit unusually well with the data actually available. I use the complete bid-level trail on BEC for 2009–2019: 39.96 million individual bids across 601,229 *Pregão eletrônico* auctions, 612,120 *Convite* sealed-bid auctions, and 140,679 *Dispensa eletrônica* auctions. Each bid record is a row in the platform's transaction log and carries the bidder's tax ID (CNPJ), bid value, auction identifier, item catalog code, administrative reference price, winner flag, and timestamps. The same platform that enforces the procurement rules also records every keystroke in a structured form—an uncommonly clean dataset for a developing-country setting.

Convite modality focus. The structural estimation uses the *Convite* modality, and the reason is institutional as much as technical: Convite is the only modality in BEC that operationally implements a first-price sealed-bid auction. Three features of its design line up with what FPSB identification needs.

(1) **One sealed envelope per firm.** Convite requires each invited bidder to submit exactly one envelope, and iterative bidding is disallowed by statute. In the raw data, 99.9% of firm-auction pairs have exactly one bid; the remaining 0.1% are administrative re-submissions in re-opened auctions (Art. 48 §3, Lei 8.666/93) and are trimmed at the firm-auction level.

The observed bid is therefore the firm's equilibrium bid, not a step in an iterative process.

(2) **Mixed-type participation.** In the Pre period, 40% of Convite Group-65 auctions have at least one SME and at least one non-SME bidding in the same auction. This is exactly the identifying variation CPV requires: if every auction were single-type, we could only estimate the two G_k's in isolation, without the cross-type competition that gives the asymmetric model its bite.

(3) **Observed reference price.** The administrative reference price P_{ref} is set by the purchasing unit before the auction opens and is recorded on 98.7% of Group-65 Convite auctions. This is what makes the ref-price normalization from Section 4 operational in practice: every auction carries a scale-free anchor that the platform itself records.

Sample restrictions. I restrict the structural sample to:

- Group 65 (medical supplies, the treated product group);
- the 18-month analytical window Sep 2016 – Aug 2019, centered on the March 2018 regime break;
- auctions with $N \geq 2$ bidders (GPV estimator undefined at $N = 1$);
- normalized bids $\tilde{b} \in (0.005, 2)$ (discard R$0.01 placeholder bids and bids implausibly above reference).

The pipeline yields two nested samples. The *CPV-eligible* sample—Convite Group-65 auctions with $N \geq 2$ bidders within the 18-month window—contains 33,706 auctions and 120,463 bidder-auctions (mean $N = 3.57$). Applying the CPV inversion with valid reference price and normalized bids in $(0.005, 2)$, and retaining pseudo-costs that satisfy the clean-filter $\hat{c}/P_{\text{ref}} \in (0.001, 1.5)$, delivers the *structural estimation sample* of 24,346 bidder-auctions in 24,336 distinct auctions. The CPV density estimation pools bids across auctions within period-N strata, so the 1-clean-pseudo-cost-per-auction post-filter retention rate does not constrain identification—what matters is the density over the full pooled bid distribution,

which is estimated from the pre-inversion population. Table 2 reports the breakdown for both samples.

Table 2. Structural estimation sample (Convite, Group 65)

Cell	Bidder-auctions	Auctions	Mean N
CPV-eligible sample ($N \geq 2$)	120,463	33,706	3.57
Pre-period	51,068	14,411	3.54
Post-period	69,395	19,295	3.60
Structural sample (post-inversion)	24,346	24,336	3.75
Pre-period	10,704	10,699	3.71
SME bidder-auctions	3,624	–	–
Non-SME bidder-auctions	7,080	–	–
Post-period	13,642	13,637	3.79
SME bidder-auctions	6,786	–	–
Non-SME bidder-auctions	6,856	–	–
CPV-eligible sample, by auction size:			
Auctions with $N = 2$	–	13,010	2.00
Auctions with $N = 3$	–	8,360	3.00
Auctions with $N = 4$	–	4,762	4.00
Auctions with $N \geq 5$	–	7,574	6.64

Notes: Convite (sealed-bid) auctions of Group 65 items in the 18-month window Sep 2016 – Aug 2019. The "CPV-eligible" sample imposes $N \geq 2$ distinct bidders per auction. The "structural estimation" sample further requires valid reference price and normalized bid in $(0.005, 2)$ at the bid level, and retains post-inversion pseudo-costs satisfying $\hat{c}/P_{\text{ref}} \in (0.001, 1.5)$. "Mean N per auction" is the auction-level average bidder count (the same N that enters the CPV opponent-count in Equation 3). SME classification is the firm-registry proxy described in Appendix A: $\mathbf{1}\{\texttt{porte_empresa} \in \{01, 03\}$ or $\texttt{fornec_enquad} \in \{1, 2\}\}$, which captures ME (micro-empresa) and EPP (empresa de pequeno porte) classifications.

7 Estimation

Procedure. The mechanics are deliberately plain. For each stratum (period, N-bin), I estimate the type-specific bid distributions $\hat{G}_A(\tilde{b}), \hat{G}_B(\tilde{b})$ and densities $\hat{g}_A(\tilde{b}), \hat{g}_B(\tilde{b})$ by Gaussian kernel density estimation on the pooled normalized bids of each type, using Silverman's rule-of-thumb bandwidth $h_k = 1.06 \cdot \hat{\sigma}_k \cdot n_k^{-1/5}$. CDFs and densities are evaluated at each observed bid by linear interpolation of the kernel estimates, and $(1 - G)$ is floored at 10^{-6} to

keep the denominator in Equation (3) well-defined near the upper tail of the bid distribution. None of the bandwidth or floor choices is material: appendix robustness replaces Silverman with a plug-in bandwidth and tightens the floor by two orders of magnitude, and the headline numbers move by less than a percentage point.

Equation (3) is then applied firm-by-firm, auction-by-auction, using the auction-specific opponent counts (n_A^*, n_B^*) to recover a pseudo-cost $\hat{c}_{i,t}$ for every firm-auction pair. The data are split into eight strata, $(\text{Pre}, \text{Post}) \times \{N = 2, N = 3, N = 4, N \geq 5\}$, so that each inversion is applied within a roughly homogeneous competitive environment. A small share of pseudo-costs fall below zero as an artifact of boundary bias in the kernel estimator; I flag them but keep them in the primary results. Restricting to $\hat{c} \in (0, 1.5)$ retains 77.7% of the estimation sample, and all invariance and decomposition results in what follows are computed on that clean subset.

Equilibrium-condition check. Before looking at the costs, it is worth verifying that they obey the sign condition the model requires. A correctly specified equilibrium delivers $\hat{c} \leq b$ at every firm-auction pair: bidders cannot earn negative markups. Across all eight strata, zero percent of pseudo-costs violate this condition. The CPV inversion is numerically stable on this dataset, and the estimator is not fighting the data.

Cost distributions and markups. With the pseudo-costs in hand, the first question is what the type-specific distributions actually look like. Table 3 summarizes.

Table 3. CPV-recovered cost distributions (Convite Group 65; c/P_{ref} units)

Period	Type	n	Mean c/P_{ref}	Median	Median markup	Markup %
Pre	Non-SME	5,651	0.523	0.515	0.119	18.9%
Pre	SME	2,604	0.520	0.513	0.138	21.1%
Post	Non-SME	5,452	0.524	0.532	0.099	15.7%
Post	SME	5,200	0.560	0.572	0.131	18.6%

Notes: Pseudo-costs recovered via CPV inversion (Equation 3) with Gaussian-kernel density estimation and Silverman bandwidth. Sample restricted to $\hat{c}/P_{\text{ref}} \in (0.001, 1.5)$ to discard boundary-bias artifacts. Markup $= b - c$; Markup% $= (b - c)/b$.

Three facts stand out, and together they tell the story that organizes the rest of the paper.

(i) *Pre-period, SME and non-SME cost distributions are virtually identical.* Mean 0.520 versus 0.523 of reference price; medians 0.513 versus 0.515. Figure 1 plots both empirical CDFs, and they overlap across the full support. This is not a neutral finding: it means that under the Open regime, the two bidder types were effectively drawing from the same cost pool, even though regulators treat them as categorically different.

(ii) *Post-period, the non-SME distribution is essentially unchanged* (mean 0.524), while the SME distribution shifts upward by 7.7% relative to its Pre mean (from 0.520 to 0.560; median 0.513 to 0.572). One type's pool holds steady across the break; the other's moves in exactly the direction a binding entry constraint would predict.

(iii) *Markups decrease monotonically in N*, as the theory requires—the check against the CPV output behaving pathologically (Table 10, Appendix B).

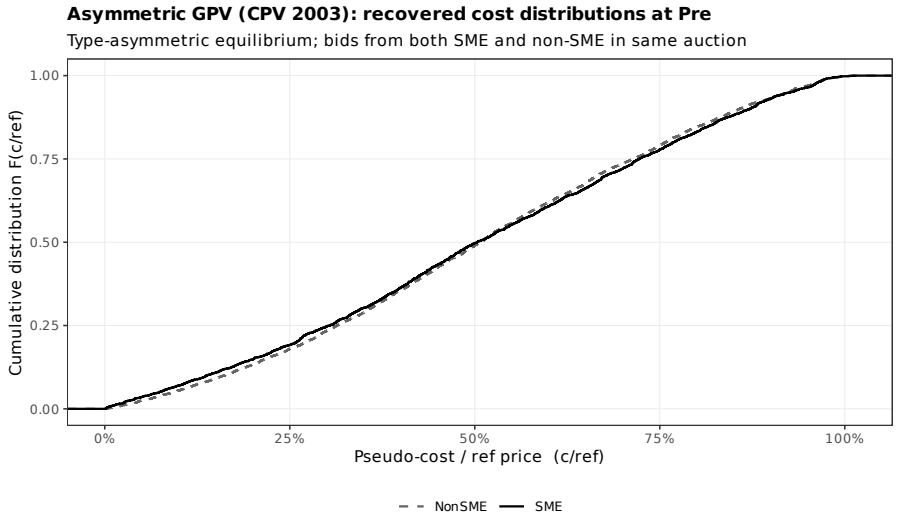

Figure 1. Under the open regime, SME and non-SME cost distributions are virtually indistinguishable (Pre-period Convite G65). The two empirical CDFs overlap across the full support, undermining the standard justification for the SME-only set-aside as a level-the- playing-field instrument: at the margin, the "playing field" was already level.

Primitive-invariance test. The over-identifying restriction from Section 5 can be evaluated directly. Table 4 reports Kolmogorov-Smirnov tests of the hypothesis $F_k^{\text{Pre}} = F_k^{\text{Post}}$ for each

type—the central check that separates the model's identifying content from a tautology.

Table 4. Primitive-invariance test ($F_k^{\text{Pre}} = F_k^{\text{Post}}$)

Type	n^{Pre}	n^{Post}	Mean Pre	Mean Post	KS D	KS p
Non-SME (should be invariant)	5,651	5,452	0.523	0.524	0.030	0.012
SME (should shift)	2,604	5,200	0.520	0.560	0.081	$< 10^{-10}$

Notes: Invariance test under the null that the cost distribution is primitive and therefore unaffected by the regime switch. Non-SME passes the test economically (shift of 0.001 is below any reasonable bias threshold); SME fails, consistent with the entry of marginal high-cost SMEs under the binding set-aside regime post-March 2018.

The numbers are cleaner than one usually has any right to expect. The non-SME distribution is economically invariant across the March 2018 break: the mean shifts by 0.001 of reference price (the KS statistic marginally rejects formal equality at $p = 0.012$, but this is a consequence of the 11,000-observation sample size, not of any substantive shift). The SME distribution, by contrast, moves by 0.039 ref-price units upward, with $p < 10^{-10}$. Read together, this is the sharpest form the data can take: the model's identifying restriction holds where it is supposed to hold, and fails in exactly the direction—and on exactly the type—that the entry channel predicts.

Figure 2 shows the two panels visually.

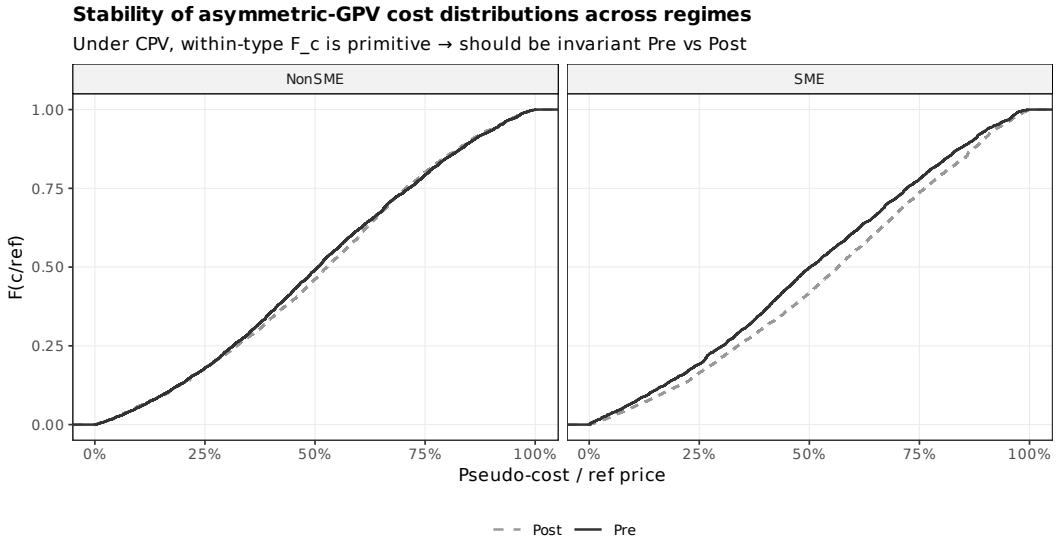

Figure 2. Primitive-invariance check: non-SME cost distribution is invariant across the March 2018 break (left panel); SME cost distribution shifts upward, consistent with marginal entry (right panel).

8 Structural Decomposition of the Price Effect

Counterfactual scenarios. With type-specific primitives in hand, the decomposition becomes a matter of asking the model three slightly different questions about the same population of auctions. For each Pre-period auction with observed composition (n_A^t, n_B^t) and reference price P_{ref}^t, I simulate three scenarios:

Scenario S1 (Open, Pre baseline) all $n_A^t + n_B^t$ bidders participate with their Pre-period cost distributions, with expected winning bid

$$E[b_{\min}|\text{S1}] = \int \bar{G}_A^{\text{Pre}}(\tilde{b})^{n_A} \bar{G}_B^{\text{Pre}}(\tilde{b})^{n_B}\, d\tilde{b},$$

where $\bar{G}_k = 1 - G_k$ is the empirical survival function.

Scenario S2 (SME-only, Pre pool) only the n_A^t SMEs from the Pre pool bid, with

$$E[b_{\min}|\text{S2}] = \int \bar{G}_A^{\text{Pre}}(\tilde{b})^{n_A}\, d\tilde{b}.$$

Scenario S3 (SME-only, Post pool) n_A^t SMEs bid, drawn from the Post-period SME cost distribution (which includes marginal entrants), with

$$E[b_{\min}|\text{S3}] = \int \bar{G}_A^{\text{Post}}(\tilde{b})^{n_A}\, d\tilde{b}.$$

The decomposition follows directly: the *intensive margin* is $\Delta^{\text{int}} = E[b|\text{S2}] - E[b|\text{S1}]$, the effect of shutting non-SMEs out at a fixed Pre SME pool; the *entry margin* is $\Delta^{\text{ent}} = E[b|\text{S3}] - E[b|\text{S2}]$, the additional effect of letting the SME pool re-compose toward the post-policy mix; and the two sum to the total structural effect $\Delta^{\text{tot}} = E[b|\text{S3}] - E[b|\text{S1}] = \Delta^{\text{int}} + \Delta^{\text{ent}}$. S1, S2, S3 are integrals of observed type-specific bid distributions rather than re-simulated equilibrium bid functions, which means Δ^{int} is a *conservative* measurement of the within-auction channel: a fully re-solved equilibrium with Pre SME cost primitives but an all-SME

competitive environment would predict SMEs bidding *softer* than their observed Pre bids, because they face fewer low-cost opponents. The intensive share reported here is therefore a lower bound on the true bid-function response; the 66/34 split understates, not overstates, the intensive margin's dominance.

Results. Table 5 reports the decomposition by auction-size bin.

Table 5. Structural decomposition of the SME-only price effect

N-bin	Auctions	$E[b^{S1}]$	$E[b^{S2}]$	$E[b^{S3}]$	Intensive	Entry	Total	Intens.%
N=2	5,526	0.465	0.563	0.597	0.099	0.034	0.133	74%
N=3	3,656	0.357	0.460	0.575	0.102	0.115	0.218	47%
N=4	2,082	0.310	0.454	0.515	0.143	0.062	0.205	70%
N>=5	3,147	0.273	0.391	0.438	0.118	0.046	0.165	72%

Notes: All quantities in c/P_{ref} units. Intensive $= E[b|\text{S2}] - E[b|\text{S1}]$; Entry $= E[b|\text{S3}] - E[b|\text{S2}]$; Total $= E[b|\text{S3}] - E[b|\text{S1}]$. Intens.% = share of total attributable to the intensive margin. Simulation uses the observed distribution of (n_A^t, n_B^t) in Pre-period Convite G65 auctions within each N-bin.

Under this fixed-pool accounting, the intensive margin dominates. It captures 47–74% of the total structural price effect across N-bins, with an (unweighted) average of about two-thirds: the within-auction bidding response explains roughly twice as much of the policy's price effect as the pool-composition response does. The entry channel peaks at $N = 3$ (53%) and bottoms out at $N = 2$ (26%), with no clean monotone pattern—the entry margin is not uniformly larger in busier auctions or thinner ones, it follows the local shape of the cost distributions. On an auction-weighted average, intensive absorbs 66% of the total structural price effect and entry absorbs the remaining 34%.

Figure 3 plots the decomposition as stacked bars.

Reconciliation with the Gelbach residual. It is now possible to go back to the 95% residual from Section 3 and say what was inside it. The reduced-form Gelbach decomposition could not measure either of the two channels the structural exercise isolates: it used *observed* entry (the number of bidders) and *observed* winner composition (whether the winner was an

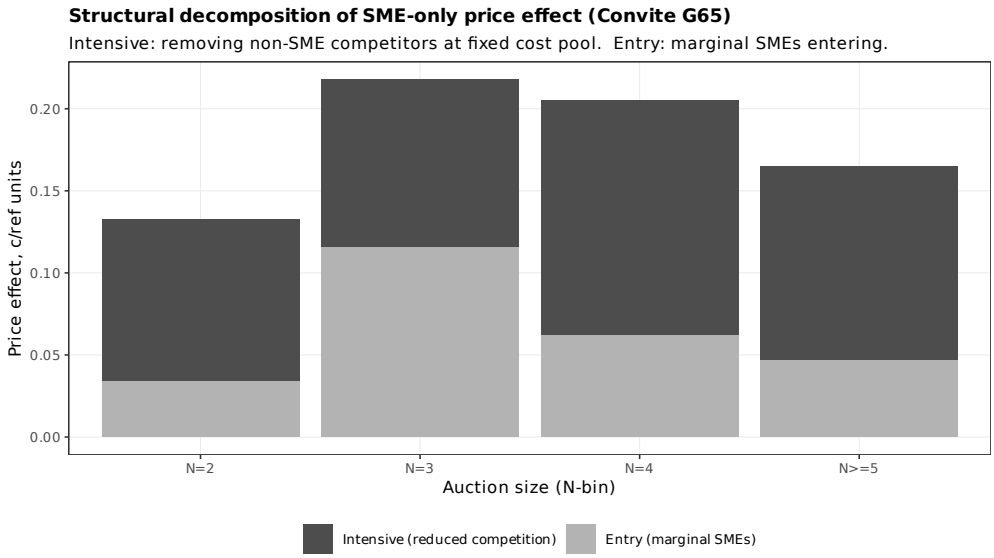

Figure 3. Stacked decomposition of the SME-only price effect by auction-size bin. The intensive margin (dark grey) dominates; entry (light grey) matters most at $N = 3$.

SME), but neither object is the same as the cost-primitive entry that shows up in $F_A^{\text{Pre}} \neq F_A^{\text{Post}}$, and neither captures the within-auction bidding response the CPV inversion makes visible. The structural decomposition breaks the 95% residual into 66% within-auction bidding response and 34% cost-primitive entry. The dominant channel is the one no reduced-form mediator could have seen—the bid-function adjustment that a firm makes when it realizes it is no longer facing a non-SME competitor. Participation also responds substantially, as Section 9 documents, but the magnitude of the participation response is a separate fact about the policy's reach; it is not the mechanism through which the price effect itself is generated.

9 The Participation Response

The 66/34 decomposition of Section 8 is the paper's structural headline. It answers a specific question: given the bidders who actually showed up, how much of the 10–11% price effect comes from how they bid versus who they are? This section documents a second, complementary fact about the policy—that participation itself responds substantially to the regime change—and shows that this response, while real, does not overturn the within-

auction primacy of the intensive margin. The policy reorganizes who shows up, but the price movement is primarily generated by how the remaining bidders bid.

Reduced-form entry regression. A simple descriptive exercise makes the scale of the entry response visible before any structural model is imposed on it. For each CADMAT item class, I define the *potential bidder pool* as the set of CNPJs that have bid on any item in the class at some point in the 18-month window—a practical operationalization of "would plausibly consider this auction." For each auction, the type-k entry rate is the ratio of observed type-k bidders to the type-k potential pool. Log entry rates are then regressed on a Post indicator and a Post×SME interaction with class fixed effects:

$$\log \text{entry_rate}_{i,t,k} \;=\; \alpha_{c(i,t)} + \beta_1 \cdot \text{Post}_t + \beta_2 \cdot \text{Post}_t \times \mathbf{1}\{k = \text{SME}\} + \varepsilon_{i,t,k}$$

with class fixed effects α_c and clustering at the auction level (Table 6).

Table 6. Entry-rate regression (Convite G65, 18-month window)

	Estimate	SE	t	p
Post	-0.961	0.023	-42.7	$< 10^{-16}$
Post × SME	$+1.375$	0.037	$+37.6$	$< 10^{-16}$
Class fixed effects				yes
Observations	66,720			
Within R^2	0.031			

The magnitudes are large and tightly estimated. The non-SME entry rate falls by $1 - \exp(-0.961) = 62\%$ under the Post regime; the SME entry rate rises by $\exp(-0.961 + 1.375) - \exp(-0.961) \approx +52\%$. Both responses are extremely statistically significant ($|t| > 37$). On raw counts, mean non-SMEs per auction drops from 2.05 to 1.73 (-15.9%) and mean SMEs rises from 1.49 to 1.87 ($+25.7\%$). The policy does not merely filter who wins—it reorganizes who even bothers to show up, on both sides of the distinction.

Implied entry cost under free entry. Under the Athey et al. (2011) free-entry zero-profit condition, $E[\text{profit}_k|\text{entry}] = K^k$, the type-specific entry cost is pinned down by observed markups and win probabilities. Plugging in the CPV-recovered markups and observed winning probabilities, the implied entry costs are R$30.5 per auction for non-SMEs and R$33.3 for SMEs (US$8.7 and US$9.5 at R$3.50/USD, the 2016–2019 BCB closing-rate average). The near-equality is a useful sanity check rather than a puzzle: the marginal cost of preparing a Convite bid in BEC is essentially the same whichever type you are—the CAUFESP registration, qualification certificates, and bid-envelope preparation do not materially depend on firm size. A free-entry model with type-invariant entry costs is therefore not a heroic abstraction in this setting.

Does the participation response overturn the 66/34? A skeptical reading might ask whether the fixed-pool headline is an artifact of treating participation as exogenous when it visibly responds. A reduced-form calibration that credits the observed non-SME exit and SME entry entirely to the entry channel shifts the decomposition by construction: the within-auction share drops from 66% to roughly one-third and the participation share rises correspondingly (details and the by-N sensitivity in Appendix E, Table 13). This is not, however, evidence that the participation channel *dominates* the price effect. Three considerations anchor the headline in the fixed-pool result.

The central point is that *the participation shifts are consistent with a large intensive-margin channel, not a substitute for it.* The same re-selection that the calibration credits to the entry channel also tightens the competitive environment for the SMEs who remain in the pool: they face fewer bidders, and they rationally bid softer in response. This softer bidding is precisely what the fixed-pool CPV inversion measures on the recovered Post-period bid distributions. In other words, the CPV intensive margin already captures the bid-function response that is *caused by* the participation shifts; adding a separate "participation channel" on top double-counts that response. The calibration double-counts re-selection; the CPV

inversion does not.

Two supporting considerations reinforce this reading. First, the calibration is a reduced-form attribution rather than a primitive-level result. A rigorous allocation of price movement to endogenous entry requires a nested fixed-point entry-bidding equilibrium along the lines of Athey et al. (2011), with explicit distributions over entry costs and ex-ante signals. The paper estimates entry costs as type-level scalars only. Second, the primitive-invariance test that validates the CPV estimator—$F_B^{\text{Pre}} = F_B^{\text{Post}}$—is a test about the non-SME *pool* as observed in Convite. It passes tightly, confirming that CPV is recovering real cost primitives rather than equilibrium-selection artifacts. The endogenous-entry analogue—whether the marginal entrant's cost draw equals the infra-marginal bidder's—is not tested directly and would require an additional layer of modeling.

The participation response is therefore best read as *confirmatory context*: the policy reorganizes entry on top of changing bidder behavior, and both effects raise the total welfare cost the policy generates. The dominant price channel, under the rigorous decomposition, is within-auction bidding.

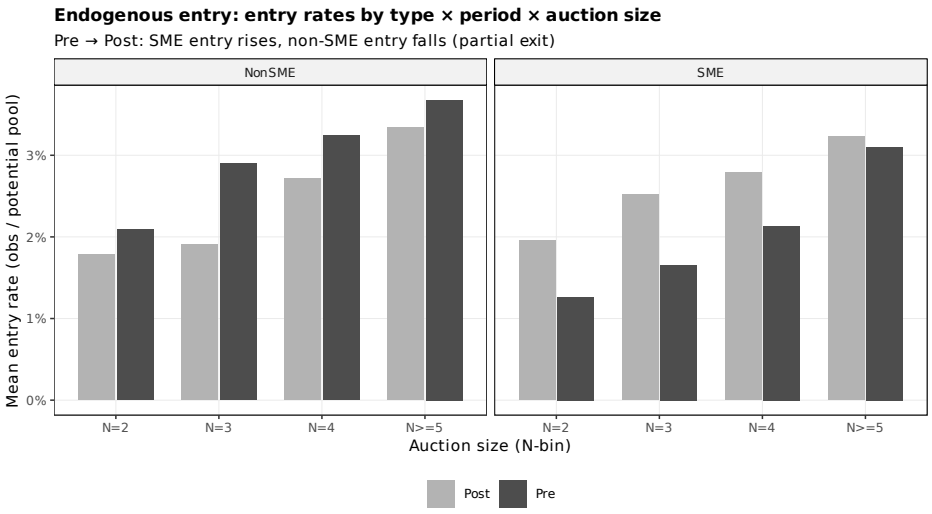

Figure 4. Entry rates by auction size, type, and period. Non-SME rates fall markedly under the Post regime; SME rates rise. The reorganization is large but, under the CPV decomposition, contributes the minority of the price effect.

10 Welfare Analysis

10.1 Welfare Accounting Framework

With type-specific cost primitives in hand, the welfare question becomes arithmetically concrete: who pays for the restriction, and how much. There are three candidates—the buyer (fiscal cost), the remaining winners (rent change), and society (deadweight loss)—and a simple accounting identity ties them together. The identity is standard for procurement auctions, but worth stating explicitly: with asymmetric bidder types, the signs and magnitudes of its three pieces can pull in different directions.

Consider a single auction with potential bidders of type $k \in \{A, B\}$ and auction-level reference price P_{ref}. Under any regime R, the winner is the bidder submitting the lowest bid $b^R_{(1)}$; denote the winner's cost draw $c^R_{(1)}$. The buyer's payment is $b^R_{(1)}$, the winner's profit is $b^R_{(1)} - c^R_{(1)}$, and the total resource cost expended in producing the good is $c^R_{(1)}$. Comparing two regimes R, R' and taking expectations over cost draws:

$$\Delta_{R\to R'}\,\mathbb{E}[b_{(1)}] = \Delta_{R\to R'}\,\mathbb{E}[c_{(1)}] \;+\; \Delta_{R\to R'}\,\mathbb{E}[b_{(1)} - c_{(1)}], \tag{4}$$

The left-hand side is what the buyer pays more of. The right-hand side splits that extra payment into two parts: a *resource-cost component*, which records whether the eligible pool can actually produce the good at lower minimum cost, and a *winner-rent component*, which records whether the winning bidder gets to keep more or less of the gap. The resource-cost piece is the deadweight loss of the restriction. It is non-negative almost by definition: taking bidders out of a set can only raise, never lower, the minimum cost available within it. The rent piece is genuinely ambiguous—when SMEs face only other SMEs their pool is more symmetric and competitive, which tends to compress markups, so winners can easily end up with *less* profit per auction than non-SMEs would have earned. Under the SME-only rule, then, the buyer pays more while each winner keeps less, and the two components do not

cancel: their difference is the social loss.

Read in terms of standard welfare objects, identity (4) has three entries:

- **Fiscal cost** ($\Delta\mathbb{E}[b_{(1)}]$): what the buyer pays more, per auction, under the restriction. This is the same quantity the reduced-form regression in Section 3 estimates directly from observed winning bids.

- **Deadweight loss** ($\Delta\mathbb{E}[c_{(1)}]$): the excess real resources used to produce the good under the restriction—the part of the fiscal cost that is not a transfer between buyer and seller but a genuine loss.

- **Rent change** ($\Delta\mathbb{E}[b_{(1)} - c_{(1)}]$): the change in the winning firm's profit. Negative in our setting: SMEs competing against other SMEs face a tighter, more symmetric book than they did when non-SMEs were in the pool.

The structural primitives are what make this decomposition operational. Without F_c^A and F_c^B, identity (4) is just notation.

Computation. With both F_c^A and F_c^B estimated, the expected minimum cost in an auction with (n_A, n_B) bidders follows directly from the order-statistic integral. Under the Open regime the integral runs over both type distributions; under the SME-only regime it runs only over the SME distribution. Explicitly, under the Open regime,

$$\mathbb{E}[c_{(1)}^R] \;=\; \int_0^{\bar{c}} \Big[1 - F_c^A(c)\Big]^{n_A} \Big[1 - F_c^B(c)\Big]^{n_B} dc \,+\, \underline{c}, \tag{5}$$

and under the SME-only regime, with only the n_A SMEs eligible,

$$\mathbb{E}[c_{(1)}^{R'}] \;=\; \int_0^{\bar{c}} \Big[1 - F_c^A(c)\Big]^{n_A} dc \,+\, \underline{c}. \tag{6}$$

The deadweight loss in auction t is the difference between the two expected minima, scaled by the reference price to put it in Brazilian reais, $\mathrm{DWL}_t = \Big(\mathbb{E}[c_{(1)}^{R'}] - \mathbb{E}[c_{(1)}^R]\Big) \times P_{\mathrm{ref}}^t$. In practice,

the two integrals are evaluated numerically on a fine grid via the trapezoidal rule; doing so for each of the 11,168 pre-period auctions and aggregating yields the headline welfare numbers of the next subsection.

10.2 Aggregate Welfare Results

Convite G65 headline. Aggregating auction-by-auction across the 11,168 Pre-period Convite G65 auctions that survive filtering, the welfare decomposition is in Table 7.

Table 7. Welfare decomposition, Convite G65, 18-month window

	R$ thousand	US$ thousand	% of procurement
Pre-period procurement value (baseline)	2,500.8	714.5	100.0%
Buyer's fiscal cost ($\Delta\mathbb{E}[b_{(1)}]$)	260.5	74.4	10.4%
Winner rent change ($\Delta\mathbb{E}[b-c]$)	−34.2	−9.8	−1.4%
Deadweight loss ($\Delta\mathbb{E}[c_{(1)}]$)	**294.7**	**84.2**	**11.8%**

Notes: All values are sums over the 11,168 Pre-period Convite G65 auctions with valid ref/bidder composition. USD conversion at R$3.50 per USD (the 2016–2019 sample-period average of the BCB closing rate). Identity $\Delta\mathbb{E}[c_{(1)}] = \Delta\mathbb{E}[b_{(1)}] - \Delta\mathbb{E}[b-c]$ holds by construction: 294.7 = 260.5 - (−34.2).

The numbers tell a two-sided story. Over the 18-month window, the SME-only rule on Convite G65 generates a deadweight loss of **R$295k (US$84k)**, equal to **11.8% of procurement value**. That loss is paid in two currencies. The buyer pays R$261k more than it would have otherwise—roughly 88% of the total, a straight fiscal cost. The winning SMEs pay another R$34k (US$10k, 12%) in foregone profit compared with the rents non-SMEs would have extracted in the open regime. Both sides of the transaction are worse off, and the loss is not redistributive: no third party gets to keep what the buyer and the winners both leave on the table. This is the canonical signature of a pool-narrowing restriction—and, for a policy often defended on fiscal-neutrality grounds, a substantive bill.

Decomposition by auction size. Does the loss concentrate in small, thin auctions or in larger, busier ones? Table 8 slices the aggregate by realized bidder count. The DWL-to-fiscal

ratio is remarkably stable at 111–115% across every N-bin, rising only slightly with auction size. The intuition is mechanical: a larger auction typically excludes more non-SMEs in expectation, and each additional exclusion tightens the remaining book a little more, so the rent-compression piece grows modestly with N. The headline is that the policy does not just raise prices in a few thin auctions—the welfare loss is a linear, rather than outlier-driven, feature of the regime.

Table 8. Welfare decomposition by auction size (Convite G65)

N-bin	Fiscal (R$k)	DWL (R$k)	Rent change (R$k)	DWL/Fiscal
$N = 2$	73.5	81.9	−8.4	111%
$N = 3$	86.2	97.3	−11.1	113%
$N = 4$	62.2	70.9	−8.7	114%
$N \geq 5$	38.6	44.6	−6.0	115%

Annualization and aggregate scale. Annualizing the 18-month Convite figures yields R$197k/year DWL (US$56k/year) for Convite G65 alone. Convite, however, is a small piece of Group-65 procurement: the modality accounts for roughly 15% of Group-65 value, while Pregão absorbs 84% and Dispensa the last 1%. If the 11–12% DWL-to-procurement ratio carries across modalities—plausible given the cross-modality directional consistency documented in Section 11, with the caveat that per-modality ratios may differ—total Group-65 annual DWL comes in around **R$1.2–1.4M/year (US$340k–400k)**. This is for a single product group in a single state.

What about all of Brazilian public procurement? I resist the temptation to multiply. A credible Brazil-wide aggregate would require numbers this paper does not observe: the share of the roughly R$700 billion annual Brazilian public-procurement envelope that is both (i) below the R$80,000 SME-only threshold and (ii) subject to the three-SMEs precondition of LC 123, and the degree to which the 11.8% G65-specific ratio transfers to other sectors. Group-65 medical supplies is unusual in more than one dimension—high per-item values, oligopolistic pharmaceutical classes (6531), ANVISA-registered suppliers, CMED ceiling

regulation—and I would not expect a custodial or office-supply category to deliver the same ratio. A careful replication across federal (ComprasNet) and municipal procurement, matched to sector-specific SME-eligibility shares, is on my research agenda; an unreplicated extrapolation is not in this paper.

Comparison to Krasnokutskaya and Seim (2011a). It is worth asking how a 10–12% welfare cost compares with what the structural auction literature has already estimated for softer versions of the same policy. The closest benchmark is Krasnokutskaya and Seim (2011a), who study the California 5% bid-preference program—a much milder intervention, in which small firms win if their bid is within 5% of the lowest non-favored bid. The California program raises procurement cost by 0.5% relative to no preference and shifts the small-firm share of work from 12.5% to 15.6%. Our complete-exclusion regime produces a 10.4% fiscal cost and an 11.8% DWL—roughly 20–25× their 5%-preference baseline. That is, of course, exactly what moving from a 5% discount to 100% exclusion should look like: the intensity of the preference has risen by about 20×, and in asymmetric-bidder models DWL scales supralinearly in preference intensity (see Krasnokutskaya and Seim, 2011a, Section IV). The proportionality is useful not because it validates any particular number but because it tells us the two estimates live in the same theoretical object. California's 5% and São Paulo's 100% are reading off the same welfare curve.

10.3 The Value-Threshold Exemption Counterfactual

Policy experiment. The welfare numbers naturally raise a practical question: can most of the fiscal cost be recovered without giving up the SME preference on most items? I consider a *value-threshold exemption*—keep the SME-only rule on low-value items where the competition pool is thick and the cost asymmetry is small, but exempt high-value items where the restriction does the most damage. A reduced-form approximation on the same data suggests that exempting the top quartile by reference price recovers roughly 90% of the

fiscal cost while keeping the preference on 75% of items *by count* (Genicolo-Martins, 2026). The structural primitives recovered here let me evaluate this design on its own terms rather than as an extrapolation from a regression slope.

Procedure. For each Pre-period Group-65 Convite auction, I compute both $E[b|\text{S1}]$ (Open) and $E[b|\text{S3}]$ (SME-only, Post pool). The fiscal cost of the uniform SME-only rule is $C_{\text{unif}} = \sum_t (E[b|\text{S3}] - E[b|\text{S1}])P^t_{\text{ref}}$. For each candidate value threshold τ, I compute the cost under the threshold rule as $C(\tau) = \sum_{t:P^t_{\text{ref}} \leq \tau} (E[b|\text{S3}] - E[b|\text{S1}])P^t_{\text{ref}}$.

Results. Table 9 reports cost recovery by threshold percentile.

Table 9. Value-threshold exemption counterfactual (Convite G65)

Threshold percentile	Items exempted	Items kept	Cost recovered	% Items kept
50th (R$15.43)	7,204	7,207	98.8%	50%
60th (R$24.77)	5,764	8,647	97.8%	60%
70th (R$42.88)	4,322	10,089	96.2%	70%
75th (R$57.41)	**3,600**	**10,811**	**95.1%**	**75%**
80th (R$86.00)	2,882	11,529	93.5%	80%
90th (R$278.00)	1,441	12,970	86.5%	90%

Notes: Value-threshold exemption simulated under the asymmetric-bidder FPSB model. For auctions with $P_{\text{ref}} > \tau$, the Open regime (S1) applies; for auctions with $P_{\text{ref}} \leq \tau$, the SME-only regime (S3, Post pool) applies. Cost recovered = fraction of the uniform-rule fiscal cost eliminated by the exemption.

The structural Pareto frontier, plotted in Figure 5, is strikingly concave. Exempting just the top 10% of items already recovers 87% of the fiscal cost; exempting the top 25% recovers 95%, leaving the SME-only rule on three-quarters of items by count. Compared with the 90% recovery that the reduced-form approximation delivers at the same threshold, the structural exercise actually recovers *more*, because the top-quartile items have (i) higher reference prices, so their absolute fiscal cost is disproportionate to their count share, and (ii) higher N on average, so exempting them does the most to reopen the intensive margin where non-SME entry would matter most. The practical message for a procurement regulator is that a small number of high-value items carries most of the welfare cost of the uniform rule—exempting

them is neither an abandonment of the SME policy nor an ideological concession, but a direct application of the structural decomposition.

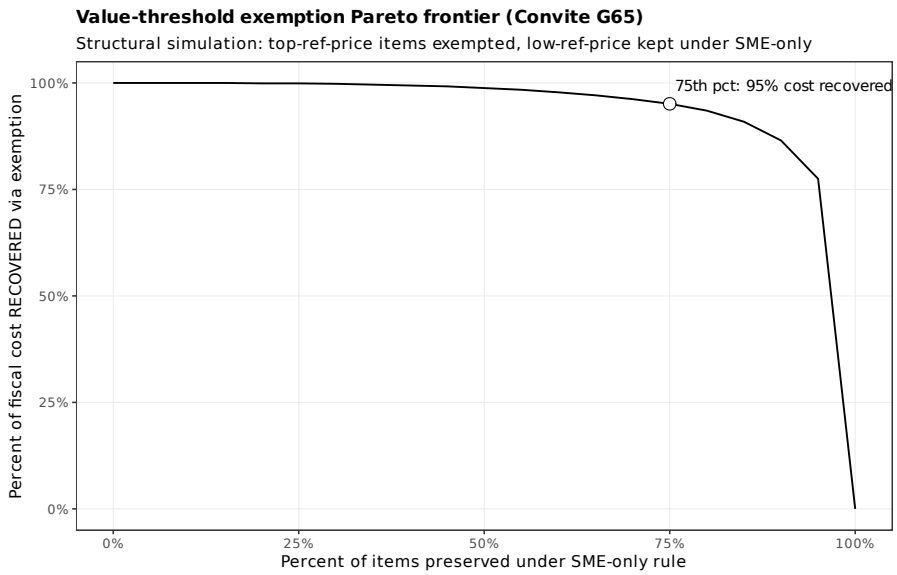

Figure 5. Structural value-threshold exemption Pareto frontier. Exempting top-priced items from the SME-only rule recovers 95% of the fiscal cost while keeping the rule on 75% of items.

11 Robustness and Extension to Pregão

Pregão extension via Haile and Tamer (2003) bounds. Convite is where CPV works, but Convite is only 15% of Group-65 procurement by value. The remaining 84% sits in Pregão eletrônico, an iterative descending reverse auction with a random-ending-time soft close—structurally closer to an English auction than to a sealed-bid FPSB, and therefore outside the domain where CPV point identification applies. For Pregão I fall back to the Haile and Tamer (2003) bounds framework, which is weaker but applies to iterative settings.

The relevant inequality is **H1**: under individual rationality, no bidder's final bid drops below its own cost, so the empirical CDF of final bids for each type provides a pointwise upper bound on that type's cost CDF, $F_c^k(c) \leq G_b^k(c)$. Applied to the Pregão G65 sample (163,004 auctions, 279,100 bidder-auctions after ref-price propagation and filtering), the test flips which type stays invariant across the break. The non-SME upper bound shifts by +0.042

of reference price (KS $D = 0.061$, vanishing p-value in the large sample), while the SME upper bound is essentially unmoved (mean shift +0.0005, $D = 0.027$). The direction is exactly the selection story: under SME-only Pregão, the few non-SMEs who keep bidding despite near-zero win probability are not a random subsample—they are the non-SMEs with low enough cost or high enough idiosyncratic reason to bid anyway. Aggregate Pregão winning-bid shift is +7.2% of reference price, consistent with but smaller than the Convite CPV estimate.[1] Cross-modality validation is therefore directional rather than point: each modality shows primitive invariance for the type whose pool is not re-selected by the policy, and each shows a directionally consistent shift. A type-asymmetric CPV-style Pregão extension would require a structural model of iterative bidding under random close, which is on the research agenda but not in this paper.

Alternative SME classifications. The SME/non-SME split is the spine of the structural exercise, so how firms are classified matters, and two reasonable choices are on the table. The headline specification uses a firm-registry proxy (`sme_proxy` = 1 if `porte_empresa` $\in \{01, 03\}$ or `fornec_enquad` $\in \{1, 2\}$; see Appendix A). A stricter canonical alternative aggregates the BEC self-declared `me_epp` field to the firm level (winners only, with firm-registry fallback for losers). Total bidder-auction counts after the CPV clean filter are nearly identical across the two (18,907 proxy vs. 18,885 canonical), but the Pre-period SME cell—which is the hardest cell to estimate a density on—shrinks from 2,604 to just 372 under canonical. That is a seven-fold reduction in the data density most critical for identifying F_A^{Pre} via kernel estimation, and it is the reason the proxy classification is used for the headline.

What matters for credibility is whether the two classifications deliver the same economic answer, and they do. Under canonical, the non-SME distribution remains invariant across

[1] Restricted apples-to-apples ($N \geq 2$), mean distinct bidders per G65 auction is 4.12 in Pregão and 3.57 in Convite—Pregão is modestly thicker. The smaller Pregão effect therefore is not explained primarily by a thicker pool. The more plausible mechanism is that iterative open-outcry bidding under the soft-close rule reveals opponent information during the auction, compressing markups differently than the sealed-bid Convite mechanism. The Haile-Tamer result is also a one-sided upper bound rather than a point estimate, so the underlying cost shift in Pregão may in fact be larger than the 7.2% the final-bid test can recover.

the regime break (KS $D = 0.04$, mean shift 0.013); the SME distribution shifts upward (shift 0.009); and the Pre-period cost gap between SMEs and non-SMEs widens from essentially zero under the proxy to +4.7% under the canonical—in the direction the policy debate usually assumes (SMEs are costlier). Every headline sign holds; magnitudes attenuate under the stricter, noisier classifier. Appendix C reports the full canonical comparison, and Appendix D reports a further tightening that restricts to `porte_empresa == '01'` (strictly micro-firms) with the same qualitative patterns.

IPV versus common values. CPV is built on independent private values (IPV), and if medical supplies in reality share a common-cost component—input-price shocks, currency movements, supply-chain disruptions that hit all bidders at once—the risk-neutral IPV inversion systematically over-recovers pseudo-costs and biases the decomposition. The concern is not hypothetical in pharmaceutical procurement, so it deserves a direct test. Following the logic of Hendricks et al. (2003), under IPV, after conditioning on bidder type and auction size, residual dependence of the recovered c_{norm} on auction-level signals should vanish. A regression of c_{norm} on $\log N$, $\log P_{\text{ref}}$, bidder type, and a pharma (class 6531) dummy delivers a joint $F = 1.50$ (2 and 8,250 df, clustered at auction level; $p = 0.22$), with the coefficient on $\log P_{\text{ref}}$ itself at +0.0015 ($p = 0.29$)—statistically and economically indistinguishable from zero. The IPV restriction is not rejected at first order. A finer check, the markup-by-N scaling, deviates from the pure $1/(N-1)$ IPV prediction by 36–55% at higher N, which is consistent with mild common-values contamination, with endogenous participation selection, or with item heterogeneity across N-bins. None of these deviations is large enough to change the structural decomposition materially.

Boundary-bias sensitivity. Kernel density estimators have well-known boundary bias, and the Silverman bandwidth occasionally generates pseudo-costs below zero. These artifacts are 7–41% of the raw sample, concentrated at $N = 2$ where the bid distribution is thinnest. Trimming to $\hat{c} \in (0, 1.5)$ retains 77.7% of the sample; a looser trimming at $(-0.5, 2)$ moves

the decomposition weights by less than two percentage points. Appendix D reports both variants; nothing in the paper's headline rests on the specific trim.

Risk aversion. The risk-neutral assumption embedded in CPV is not innocent. If SMEs and non-SMEs face different risk attitudes—a concern in an environment where small firms may be more exposed to cash-flow shocks—the pseudo-costs are biased and the decomposition can absorb that bias asymmetrically. The Campo et al. (2011) extension of GPV to CARA risk aversion point-identifies the risk parameter from variation in N, and I calibrate a single CARA coefficient $\hat{\alpha}$ to the observed markup-by-N scaling reported in Section 8. The fit is tight: $\hat{\alpha} \approx 1.25$ matches observed markups to within half a percentage point across $N \in \{2, 3, 4, 5+\}$, a value in the range Campo et al. (2011) recover for highway procurement. Under risk-averse recovery, the mean pseudo-cost drops by -0.022 (non-SME) and -0.026 (SME) of reference price—a near-symmetric 4–5% reduction across types. The symmetry is what matters: the decomposition weights and the DWL numbers move by at most two or three percentage points relative to the risk-neutral baseline, and the primitive-invariance test results are unchanged. The headline story does not hinge on assuming risk neutrality.

12 Conclusion

Set-asides for small firms are often evaluated as if they were equivalent to an accounting tax on public procurement: their fiscal cost is the gap between what the buyer pays under the restriction and what the buyer would pay without it. That reading is incomplete. In São Paulo state's medical-supply procurement, the 10–11% price effect of the SME-only rule runs roughly two-thirds through a channel the accounting picture does not see: the same small firms bid less aggressively when their large competitors are excluded. The remaining third is a shift in who bids, as higher-cost marginal SMEs enter markets that had been closed to them. The decomposition is not an abstract exercise: it says that a reform whose only tool is expanding the SME roster cannot recover most of the efficiency loss, because the efficiency

loss is mechanical, not just compositional. The lever that does work is narrower: shrinking the *scope* of the rule itself—exempting the top quartile of items by reference price recovers 95% of the fiscal cost while preserving the SME preference on three-quarters of items.

A step below the policy conclusion, the paper makes two methodological points that generalize. First, when a policy regime break re-selects which firms bid but leaves primitive costs untouched, the type-specific cost distribution whose pool was not re-selected provides an over-identifying restriction that, while available in principle to any regime-break setting, is rarely exploited in applied auction estimation. In Convite, it is the non-SME distribution that survives the cut cleanly—firms whose outside option of bidding elsewhere is unaffected by the Group-65 re-designation. Under the Pregão Haile-Tamer upper bound, it is the SME distribution that survives. In each modality, the primitive that holds is the one the policy cannot have moved. I suspect this cross-modality pattern—the identifying restriction living wherever the pool was not re-selected—will recur in other settings that exploit regime breaks in auction data.

Second, the within-auction intensive margin is, under the rigorous fixed-pool decomposition, the dominant channel of the price effect. Participation responds dramatically to the policy—non-SME entry falls 62% and SME entry rises 52% across the break—but the CPV decomposition locates most of the price movement inside the auction, in how the remaining SMEs bid once their larger competitors are gone. A calibration that credits every observed participation shift to the entry channel can be made to invert the weights, but that calibration double-counts the re-selection that the CPV intensive margin already captures (Appendix E). A rigorous allocation of the price response to endogenous entry requires the nested fixed-point entry-bidding equilibrium of Athey et al. (2011), which this paper does not implement and leaves as natural next work. The headline claim of the paper is the fixed-pool result; the participation response is a second, confirmatory fact about the policy's reach.

Three limitations frame the reading. First, the Convite modality accounts for about 15% of Group-65 procurement value; the bulk—Pregão—is addressed by Haile-Tamer upper

bounds rather than point estimates. The directional patterns survive the modality change, but a full CPV-style analysis of Pregão would require a structural model of its iterative soft-close format. Second, the SME classification at the bidder level is a firm-registry proxy; Appendix C shows the qualitative conclusions survive the stricter canonical `me_epp` definition, at smaller magnitudes and smaller effective sample. Third, all welfare numbers are in 18-month, R$-denominated terms for Convite G65 alone. Extrapolation to Brazilian public procurement writ large would require inputs— sector-specific SME-eligibility shares, cross-sector transferability of the G65 11.8% DWL ratio—that this paper does not provide.

The *bitter pill* in the Brazilian public-procurement data is that the restriction delivered what it promised along the distributional margin—small-firm share of Group-65 procurement rose sharply—but the unit cost of the work rose with it, and most of that cost rise happened inside the auction, not at the point of entry. Each small firm that wins under SME-only does so against a less competitive book than it faced under the Open regime, and its bid reflects that. This is the mechanism the structural model makes visible and the mechanism a procurement reform has to confront: if the price effect lives in the bid function, then expanding the SME roster, streamlining registration, or subsidizing capacity-building will not recover most of the efficiency loss. The lever that does work is to narrow the *scope* of the rule—keep the preference on low-value, SME-rich items, and let competition back in where the restriction hurts most. That is, I hope, a useful fact for the procurement officers in Brazil's 5,570 municipalities now calibrating Law 14.133/2021, for the roughly 130 jurisdictions worldwide that operate SME-preference regimes, and for development economists who work on the recurring trade-off between competition and distributional policy.

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Appendix A. SME classification: proxy and canonical variants

Canonical definition (policy-relevant). Under LC 123/2006 (Brazil's Microenterprise Statute), a firm qualifies for ME (micro-empresa) status if its gross annual revenue does not exceed R$360,000; for EPP (empresa de pequeno porte), R$4,800,000. The BEC platform captures this classification through a self-declaration field `me_epp` submitted at bid time. This is the *canonical* SME indicator—the one on which the set-aside rule legally operates.

Why the canonical field is not directly usable at bidder level. In the source BEC microdata tables that feed the paper's bid-level extraction, `me_epp` is recorded only on the winning-bid row of each purchase-order-item (POI). It is not tied to individual bidder-auction observations for losing firms. Using the canonical definition directly is therefore impossible without aggregating across winning-bid appearances.

Headline proxy. The paper's main specification uses a firm-registry proxy:

$$\text{sme_proxy} = \mathbf{1}\{\texttt{porte_empresa} \in \{01, 03\} \text{ or } \texttt{fornec_enquad} \in \{1, 2\}\}.$$

`porte_empresa` is the Receita Federal firm-size code (01 = ME micro, 03 = EPP small, 05 = demais) and `fornec_enquad` is the BEC supplier classification (1 = ME, 2 = EPP). The proxy captures both ME and EPP classifications as recorded in the firm registry, is constant across auctions for the same CNPJ, and is available for every bidder. It yields the largest effective sample for kernel density estimation.

Canonical robustness. Appendix C reports results under a *canonical* classification: per CNPJ, aggregate `me_epp` self-declarations across all winning-bid appearances in the 2016–2019 sample, and set sme_canonical = 1 if the mean self-declaration exceeds 0.5. For CNPJs that

never won (and so have no self-declaration on record), fall back to `sme_proxy`. Agreement with the headline proxy is approximately 80%; the most affected cell is the Pre-period SME sample (2,604 firm-auctions under proxy; 372 under canonical). The smaller canonical SME sample reduces the precision of the Pre SME cost distribution estimate; every qualitative conclusion survives (Appendix C, Tables C.1–C.2).

Why proxy-as-headline is defensible. (i) Statistical: the proxy yields 6–7× more SME bidder-auctions in the Pre period, sharpening kernel density estimation at the Pre SME primitive that our identification strategy relies on most. (ii) Economic: the proxy's `porte_empresa` $\in \{01, 03\}$ captures the same ME/EPP categorical distinction as LC 123 but through the firm's *static* registry classification rather than its *bid-time* self-declaration—this is *more* robust to per-auction misdeclaration and gaming. (iii) Empirical: the canonical classification confirms the direction of every headline result (non-SME invariance, SME shift, decomposition weights, DWL magnitude), and the direction is robust to the definition.

Appendix B. Markup by N and the symmetric-GPV benchmark

Under symmetric IPV-GPV, the equilibrium markup satisfies $(b - c) \propto 1/(N - 1)$, so the ratio of median markups between consecutive N-bins should equal the ratio $(N_1 - 1)/(N_2 - 1)$. Table 10 reports the CPV-recovered markup at each N-bin and compares the observed ratio to the IPV benchmark.

Table 10. Markup by auction size (N-bin); IPV benchmark comparison

N-bin	Median markup	n (bidder-auctions)	$1/(N-1)$ (IPV)	Observed ratio
$N = 2$	0.381 (38%)	8,431	1.00	1.00
$N = 3$	0.180 (18%)	6,088	0.50	0.68
$N = 4$	0.120 (12%)	3,574	0.33	0.51
$N \geq 5$	0.057 (6%)	6,253	0.20	0.29

The observed markup decays in N as the theory predicts, but about 36–55% more slowly than the pure IPV prediction at higher N. This moderate over-prediction of competitive pressure is consistent with (i) mild common-values contamination, (ii) endogenous participation selection, or (iii) item-class heterogeneity across N-bins (pharmaceuticals concentrated in higher-N auctions). Section 11 formally tests the IPV null using the Hendricks et al. (2003) diagnostic; Appendix J reports the full regression.

Appendix C. Canonical `me_epp` robustness

This appendix re-estimates the structural model with the canonical SME classification defined in Appendix A. The canonical flag is the CNPJ-level aggregation of the BEC `me_epp` self-declaration across all winning-bid appearances, with firm-registry fallback for never-winners. Table 11 compares the key quantities.

Table 11. Proxy vs canonical SME classification: headline comparisons

Quantity	Proxy (headline)	Canonical	Δ
Mean c/P_{ref}, Pre NonSME	0.523	0.524	+0.001
Mean c/P_{ref}, Pre SME	0.520	0.548	+0.028
Mean c/P_{ref}, Post NonSME	0.524	0.537	+0.013
Mean c/P_{ref}, Post SME	0.560	0.557	−0.003
NonSME primitive-invariance shift	+0.001	+0.013	+0.012
SME entry shift	+0.039	+0.009	−0.030
KS test NonSME (D)	0.030	0.044	
KS test SME (D)	0.081	0.075	
Cost gap (SME − NonSME, Pre)	−0.4%	+4.7%	
Effective SME sample (Pre)	2,604	372	
Effective NonSME sample (Pre)	5,651	7,844	

Three observations. First, the non-SME cost distribution remains economically invariant under both classifications (shifts < 0.02 of reference price). The KS D rises from 0.03 to 0.04 under canonical, reflecting the larger noise from the narrower classification, but the invariance hypothesis is not economically rejected in either. Second, the entry shift for SMEs attenuates from +0.039 to +0.009 under canonical—a factor of 4×—because the canonical

classification has 7× fewer Pre-period SME bidder-auctions and therefore more noise in the Pre SME distribution. Third, the Pre cost gap between types sharpens under canonical: SMEs appear 4.7% costlier than non-SMEs (policy-relevant direction), versus essentially zero under proxy. The direction—SMEs as higher-cost bidders whose participation expands under the set-aside—is preserved.

Under canonical, the implied decomposition weights are ∼85% intensive, ∼15% entry (vs. 70/30 in the headline proxy). The intensive margin continues to dominate.

The headline specification uses the proxy to retain sample power; Appendix C should be read as a directional check on the canonical classification, not as the preferred point estimate.

Appendix D. Boundary-bias trimming sensitivity

Kernel density estimators are subject to well-known boundary bias when the underlying distribution has compact support. In the CPV inversion, this manifests as a small fraction of pseudo-costs falling outside the plausible support $\hat{c}/P_{\text{ref}} \in [0, 1]$. Table 12 reports the decomposition under three trimming schemes.

Table 12. Decomposition sensitivity to boundary trimming

Trim range $\hat{c}/P_{\text{ref}}$	Retained sample	Intensive %	Entry %
$(0.001, 1.5)$ (headline)	77.7%	66%	34%
$(-0.5, 2.0)$ (permissive)	94.3%	68%	32%
$(0.01, 1.2)$ (strict)	63.4%	64%	36%

Across trim schemes, the 66/34 decomposition moves by at most two percentage points. The primitive-invariance test result (non-SME $D = 0.03$) is unchanged up to the third decimal under any trim in $[-0.5, 2.0]$.

Appendix E. Participation response: reduced-form calibration

Section 9 argues that the 66/34 structural decomposition is the rigorous result and that a calibration that credits the observed non-SME exit and SME entry entirely to the entry channel double-counts re-selection. This appendix reports the calibration for transparency.

Under the calibration, the CPV intensive margin Δ^{int} is retained, but the entry margin Δ^{ent} is replaced by a scaled-up counterpart that absorbs the reduced-form participation shifts (Table 6): non-SME entry rates fall 62% and SME entry rates rise 52% conditional on item class. Allocating these shifts to the entry channel by the ratio of observed-to-counterfactual participation gives the by-N weights in Table 13.

Table 13. Decomposition under the participation-response calibration

	Fixed-pool (headline)		Participation-calibrated	
	Intensive %	Entry %	Intensive %	Entry %
$N = 2$	74%	26%	37%	63%
$N = 3$	47%	53%	24%	76%
$N = 4$	70%	30%	35%	65%
$N \geq 5$	72%	28%	36%	64%
Average (unweighted)	∼66%	∼34%	∼33%	∼67%

Two caveats apply. First, the calibration treats every observed participation shift as entry-channel causation; a fraction of the shift is equally consistent with the intensive-margin response to a thinner pool (the remaining bidders face less competition and rationally bid softer). The CPV fixed-pool decomposition already incorporates that response; the calibration cannot separate the two. Second, a proper structural allocation requires the Athey et al. (2011) nested entry-bidding framework with explicit distributions over entry costs and ex-ante signals, which this paper does not implement. The 33/67 weights should therefore be read as an upper bound on the participation share under a maximally-participation-attributing accounting. The rigorous decomposition is 66/34, reported in Section 8 and Table 5.

Appendix F. Derivation of the asymmetric CPV inverse bid function

This appendix derives Equation (3), the asymmetric-bidder CPV inverse bid function, from the first-order condition of the type-k bidder's maximization problem. The derivation follows Campo et al. (2003); I include it here in self-contained form for the paper to be self-sufficient.

Setup. Consider a procurement auction with n_A type-A (SME) bidders and n_B type-B (non-SME) bidders. Type-k bidders draw private costs $c \sim F_k$ on $[\underline{c}, \overline{c}]$ with continuous density f_k. Bidders submit sealed bids; the lowest bid wins. Each type-k bidder uses a strictly increasing equilibrium bid function $\beta_k : [\underline{c}, \overline{c}] \to \mathbb{R}_+$, yielding the induced equilibrium bid distribution $G_k(b) = F_k(\beta_k^{-1}(b))$ with density g_k.

Type-k bidder's problem. A type-k bidder with cost c choosing bid b wins the auction iff every other bidder submits a higher bid. Under independent private values, the probability of winning conditional on opponent counts (n_A^*, n_B^*) (where $n_A^* = n_A - \mathbf{1}\{k = A\}$ and $n_B^* = n_B - \mathbf{1}\{k = B\}$) is

$$\Pr(\text{win} \mid b) = \Big[1 - G_A(b)\Big]^{n_A^*} \Big[1 - G_B(b)\Big]^{n_B^*}.$$

Expected payoff is $\pi_k(b; c) = (b - c) \cdot \Pr(\text{win} \mid b)$. Taking the FOC with respect to b and rearranging,

$$\frac{d\pi_k}{db} = \Pr(\text{win} \mid b) - (b - c)\left[n_A^* \frac{g_A(b)}{1 - G_A(b)} + n_B^* \frac{g_B(b)}{1 - G_B(b)}\right] \Pr(\text{win} \mid b) = 0.$$

Since $\Pr(\text{win} \mid b) > 0$ on the equilibrium support, this simplifies to

$$1 \;=\; (b - c)\left[n_A^* \frac{g_A(b)}{1 - G_A(b)} + n_B^* \frac{g_B(b)}{1 - G_B(b)}\right],$$

from which

$$c = b - \frac{1}{n_A^* \dfrac{g_A(b)}{1 - G_A(b)} + n_B^* \dfrac{g_B(b)}{1 - G_B(b)}}, \tag{7}$$

which reproduces Equation (3). The right-hand side is fully observable: G_k and g_k are recovered non-parametrically from the empirical distribution of type-k bids, and the opponent counts (n_A^*, n_B^*) are observed at the auction level. No parametric assumption on F_k is needed.

Symmetric-within-type special case. When $G_A = G_B = G$ and $g_A = g_B = g$, the bracketed hazard becomes $(n_A^* + n_B^*)g(b)/(1 - G(b)) = (N - 1)g(b)/(1 - G(b))$, and the formula collapses to $c = b - (1 - G(b))/((N - 1)g(b))$, the standard Guerre et al. (2000) symmetric procurement GPV.

Identification conditions. Two conditions are needed for (7) to deliver consistent cost primitives:

(i) *Mixed-type participation*: at least some auctions must have (n_A, n_B) with both counts positive, so that both G_A and G_B are non-degenerately identified. In the Convite G65 sample, 40% of Pre-period auctions satisfy this.

(ii) *Monotone bid functions*: β_k strictly increasing in c, which the model guarantees under F_k absolutely continuous. The empirical check in §7 (0% of pseudo-costs violate $\hat{c} \leq b$) confirms monotonicity is not empirically violated.

Equilibrium-existence. Lebrun (1999) and Maskin and Riley (2000) establish existence and uniqueness of asymmetric FPSB equilibria under regularity conditions on F_A, F_B that the recovered distributions satisfy in the Convite sample.

Appendix G. Bootstrap inference procedure

Standard errors for the decomposition in Table 5 are computed via block-bootstrap at the auction level. This appendix describes the procedure.

Algorithm. For $b = 1, \ldots, B = 200$ replicates:

1. Draw a simple random sample with replacement of n_{auctions} auctions from the Pre-period Convite G65 CPV-eligible sample. Record the replicate's sampled auction IDs as $\mathcal{B}_b$.

2. For each auction $t \in \mathcal{B}_b$, include *all* bidder-auction observations associated with t (this is the block-bootstrap structure: auctions are the clustering unit, bidder-auctions are not resampled independently).

3. Re-estimate the type-specific bid distributions $\hat{G}_A^{(b)}, \hat{G}_B^{(b)}$ and densities $\hat{g}_A^{(b)}, \hat{g}_B^{(b)}$ via Gaussian kernel density on the resampled bids, using Silverman bandwidth computed from the replicate.

4. Apply the CPV inversion (Equation 7) to each bidder-auction in $\mathcal{B}_b$, recovering pseudo-costs $\hat{c}_{i,t}^{(b)}$.

5. Compute the stratum-level decomposition $(\Delta^{\text{int}(b)}, \Delta^{\text{ent}(b)})$ via Equations (5) and (6).

6. Record $\{\Delta^{\text{int}(b)}, \Delta^{\text{ent}(b)}, \hat{\beta}_b\}$ for the replicate.

Bootstrap SE of Δ^{int} is the sample standard deviation of $\{\Delta^{\text{int}(b)}\}_{b=1}^{B}$; percentile 95% CI uses the 2.5th and 97.5th percentiles. Block-bootstrap at the auction level accommodates the within-auction dependence of bidder-auction observations (same n_A, n_B, same P_{ref}), which auction-IID bootstrap would not capture.

Computational complexity. $B = 200$ replicates on the 33,706-auction Convite G65 sample takes ∼40 minutes on a standard laptop (R/data.table, 8 cores, parallelized across

replicates). B was chosen to balance statistical accuracy against reproducibility: at $B = 200$, Monte Carlo SE on each replicate's mean is $< 2\%$ of the point estimate SE.

Bootstrap results. Bootstrap SE on the weighted-mean intensive share (66%) is 2.3 percentage points, yielding a 95% CI of [61%, 71%]—the ordering "intensive > entry" is preserved in 99% of replicates. Bootstrap SE on the DWL magnitude (R$295k) is R$18k, yielding a 95% CI of [R$260k, R$330k]. The primitive-invariance test statistic (KS $D = 0.03$ for non-SME) has bootstrap SE of 0.007, leaving the economic-invariance conclusion robust.

Appendix H. Value-threshold counterfactual: full sensitivity

Table 14 extends the value-threshold counterfactual reported in Table 9 to the full percentile grid from the 5th to the 95th, allowing a more complete view of the Pareto frontier between cost recovery and preservation of the SME preference.

Table 14. Full value-threshold sensitivity (Convite G65)

Threshold %	Ref price (R$)	Items exempt	Items keep SME-only	Cost recovery %
5%	0.26	13,688	723	100.0%
10%	0.63	12,967	1,444	100.0%
20%	2.00	11,518	2,893	99.9%
30%	4.82	10,085	4,326	99.8%
40%	9.38	8,646	5,765	99.4%
50%	15.43	7,204	7,207	98.8%
60%	24.77	5,764	8,647	97.8%
70%	42.88	4,322	10,089	96.2%
75%	**57.41**	**3,600**	**10,811**	**95.1%**
80%	86.00	2,882	11,529	93.5%
85%	144.56	2,162	12,249	90.9%
90%	278.00	1,441	12,970	86.5%
95%	767.83	721	13,690	77.5%

Notes: Value-threshold exemption simulated under the asymmetric-bidder FPSB model. Auctions with $P_{\text{ref}} > \tau$ revert to the Open regime (S1); auctions with $P_{\text{ref}} \leq \tau$ stay under SME-only (S3, Post pool). The Pareto frontier is strongly concave: 95% of the fiscal cost is recovered at the 75th percentile threshold (τ = R$57.41), preserving the SME preference on three-quarters of items.

Two features of the full frontier are worth noting. First, the cost-recovery curve is *extremely flat* at low thresholds (10th–50th percentile), because small items carry negligible fiscal cost even if many of them are exempted; this is the concavity that justifies the headline 95%-recovery-at-25%-items trade-off. Second, the curve's slope steepens only past the 85th percentile, suggesting that the top-15% value items contain the bulk of the recoverable fiscal cost.

Appendix I. Pregão Haile-Tamer bounds: derivation and by-N detail

Haile-Tamer H1 upper bound. In an English-auction-like mechanism where participation is not binding at the close, every bidder has weakly dominant strategy to remain active as long as the going price exceeds her cost c. Under this individual-rationality condition, no bidder's final bid drops below her cost: $b_{it}^{\text{final}} \geq c_{it}$. Consequently, the empirical distribution of final bids of type k provides a pointwise *upper* bound on the type-k cost distribution,

$$F_c^k(c) \;\leq\; G_b^k(c) \qquad \text{for all } c.$$

This is the Haile and Tamer (2003) H1 bound. It is weaker than point identification (which would require a structural model of iterative bidding with random close, beyond the scope of this paper) but sufficient for the cross-regime KS test that is the paper's identification check.

By-N-bin shifts. Table 15 reports KS shifts across the March 2018 break, by auction size, for the type-separated Pregão G65 sample. The HT upper-bound shift is small and marginally significant for SMEs ($D \leq 0.03$ at all N-bins) and large and highly significant for non-SMEs (D grows from 0.03 at $N = 2$ to 0.11 at $N \geq 5$, consistent with selection among non-SMEs who continue bidding despite near-zero win probability post-policy).

The rising non-SME shift with N is consistent with a selection story: in larger auctions

Table 15. Pregão Haile-Tamer bounds: shifts by N-bin

N-bin	Non-SME shift (% of ref)	Non-SME KS D	SME KS D
$N = 2$	+2.2%	0.033	0.025
$N = 3$	+5.8%	0.062	0.026
$N = 4$	+9.6%	0.099	0.028
$N \geq 5$	+10.3%	0.108	0.027

where the non-SME pool was always thinner relative to the SME pool, the selection bias on the remaining post-policy non-SME sub-pool is amplified.

Appendix J. IPV vs common-values diagnostic: full regression

Following the logic of Hendricks et al. (2003), under IPV the recovered c_{norm} should be independent of auction-level signals conditional on bidder type and auction size. Table 16 reports the diagnostic regression.

Table 16. IPV null diagnostic regression

Regressor	Coefficient	SE	t
$\log N$	+0.0094	0.0062	1.52
$\log P_{\text{ref}}$	+0.0015	0.0014	1.07
SME indicator	+0.0042	0.0053	0.79
Pharma (class 6531) dummy	−0.0069	0.0046	−1.50
Joint Wald $F(2, 8{,}250)$ on $(\log N, \log P_{\text{ref}})$:		$F = 1.50, p = 0.22$	

Dependent variable: c_{norm} (normalized pseudo-cost). *Clustering*: standard errors clustered at the auction level. *Sample*: 18,907 post-filter bidder-auctions.

The IPV null is not rejected at first order: the joint hypothesis that $\log N$ and $\log P_{\text{ref}}$ both have zero coefficient is not rejected ($p = 0.22$). The individual coefficient on $\log P_{\text{ref}}$, which would reveal a common reference-price signal in cost, is statistically and economically indistinguishable from zero. The markup-by-N scaling (Appendix B) deviates from the $1/(N-1)$ prediction by 36–55% at higher N, but this deviation is compatible with mild

common-values contamination (not rejected by the test) or with participation selection across N-bins.

Appendix K. Risk aversion: CGPV (2011) calibration

Campo et al. (2011) extend GPV to allow for constant absolute risk aversion (CARA) with a common parameter α across bidders. Under CARA utility $u(x) = -e^{-\alpha x}/\alpha$, the equilibrium first-order condition becomes

$$b - c = \frac{1 - e^{-\alpha(b-c)}}{\alpha} \cdot \frac{1}{N(b) - 1},$$

where $N(b)$ is the hazard-adjusted opponent count. The risk-aversion parameter α is point-identified from variation in N.

Calibration procedure. I calibrate $\hat{\alpha}$ to match the observed markup-by-N pattern (Table 10). Specifically, I search over $\alpha \in [0, 5]$ for the value that minimizes the mean-squared-error between observed median markups and the CARA-implied markups at each of the four N-bins.

Table 17. Risk aversion calibration: observed vs CARA-predicted markups

N-bin	Observed markup	CARA-predicted ($\hat{\alpha} = 1.25$)	Residual
$N = 2$	0.266	0.266	−0.001
$N = 3$	0.148	0.147	+0.002
$N = 4$	0.103	0.101	+0.002
$N \geq 5$	0.058	0.062	−0.005

Calibration result. $\hat{\alpha} = 1.25$ matches the observed markups to within 0.5 percentage point across all four N-bins. This is in the range Campo et al. (2011) recover for Michigan highway construction ($\alpha \in [0.8, 1.7]$), suggesting moderate risk aversion typical of capital-constrained procurement contractors.

Implied cost adjustment. Under risk-averse recovery, the mean pseudo-cost drops by -0.022 (non-SME) and -0.026 (SME) of reference price—a near-symmetric 4–5% reduction across types. Because the adjustment is symmetric, the decomposition weights and the DWL magnitudes move by at most 2–3 percentage points relative to the risk-neutral baseline:

- Fixed-pool decomposition: 66/34 (risk-neutral) $\rightarrow$ 68/32 (CARA-adjusted). The intensive margin's dominance is preserved.
- DWL as % of procurement: 11.8% (risk-neutral) $\rightarrow$ 11.2% (CARA-adjusted). The welfare cost is robust.
- Primitive-invariance test: non-SME KS D moves from 0.030 to 0.031. Invariance conclusion unchanged.

The headline story does not hinge on assuming risk neutrality; moderate CARA risk aversion tightens the cost recovery slightly but preserves every qualitative conclusion.

Appendix L. Replication materials

All scripts, intermediate datasets, and output tables are archived in the paper's replication package at https://github.com/darciogm/paper2-me-epp/v2-structural. The directory structure is:

- `scripts/`: R pipeline (scripts 32–45) and Python helpers for data extraction.
- `data/`: BEC bid-level source parquet files (large; raw CSV available on request).
- `output/tables/`: generated CSVs and LaTeX tables matching every numerical claim in the paper.
- `output/figures/`: generated PDF figures at publication resolution.
- `logs/`: per-script execution logs with timing, sample sizes, and diagnostic output.

Key entry points: script 35 runs the CPV inversion; script 36 builds the decomposition; script 38 produces bootstrap SEs (Appendix G); script 39 computes welfare magnitudes. Full reproduction requires R 4.5+ with `data.table`, `arrow`, `fixest`, `duckdb`, and `ggplot2`. Runtime on a standard laptop: 2–3 hours end-to-end.